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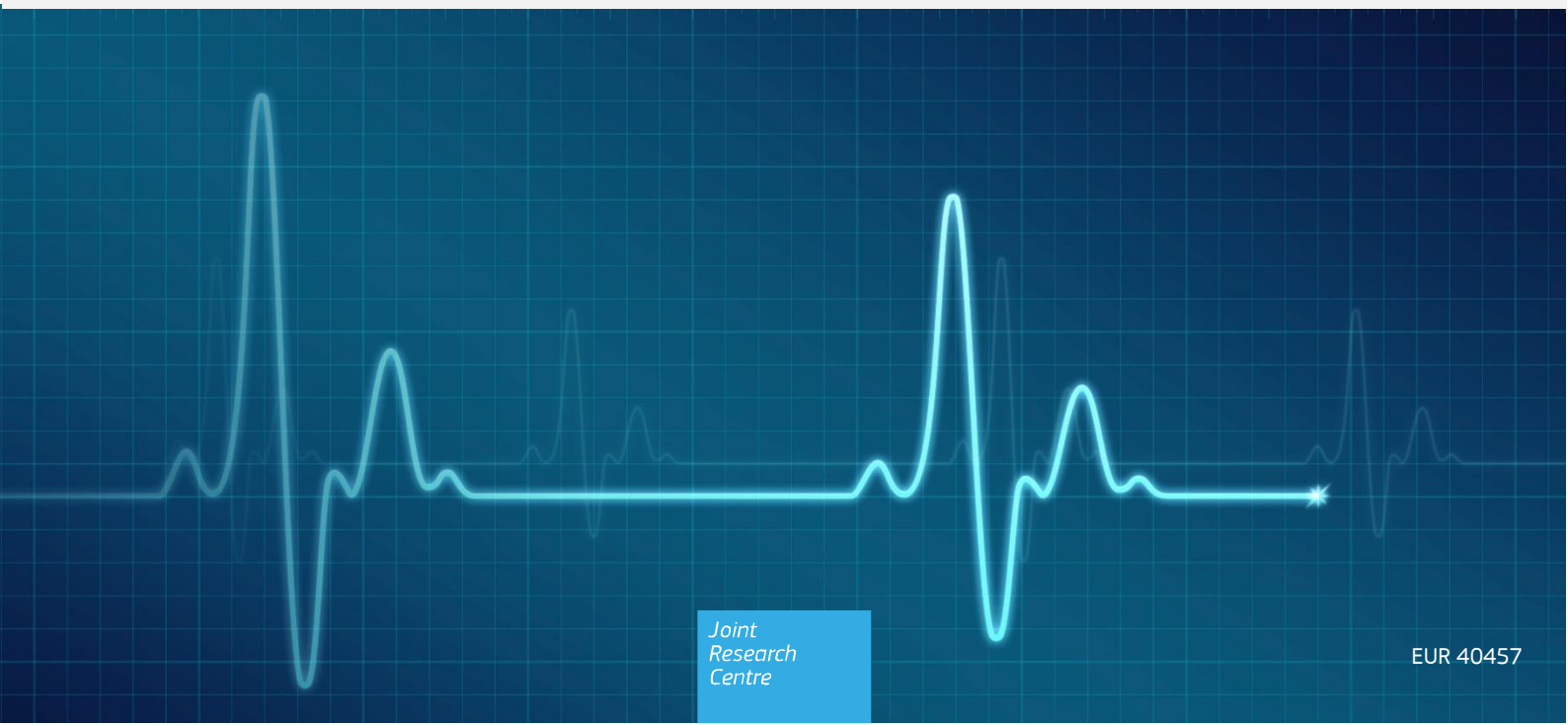
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Explainable AI in Cardiovascular Health

Methods, Applications, and Innovations

Bertolini, L., Comte, V., Ruiz Serra, V., Gierschmann, L.,
Orfei, L., Consoli, S., Ceresa, M., Querci, M., Wiesenthal, T.

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Contact information
Name: Lorenzo Bertolini
Email: Lorenzo.BERTOLINI@ec.europa.eu

EU Science Hub
https://joint-research-centre.ec.europa.eu

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Abstract

Artificial intelligence (AI) holds tremendous potential to revolutionise **healthcare** by enhancing and **supporting medical decision-making**, enabling early diagnosis, and accelerating **drug discovery processes**. While its effectiveness has been repeatedly demonstrated across various medical applications, it is essential for both patients and healthcare practitioners that humans can understand how decisions are made by AI systems, thereby increasing transparency, building trust, and ensuring clinical accountability.

This comprehensive report provides an overview of the main methodological tools developed in the literature for **AI interpretability and explainability (XAI)**, examining how these techniques function, what insights they offer into the **internal mechanisms of AI systems**, and how they establish clear relationships between inputs and outputs in medical contexts. Moreover, the report proposes an in-depth analysis of how this **interpretability tools** have been adopted in healthcare applications and, more specifically, in **cardiovascular disease (CVD)** research, exploring their primary findings, the most significant research gaps, and potential future directions for the field.

Our findings demonstrate how the research community focusing on XAI for cardiovascular disease has made a consistent and effective use of a relatively limited set of established models and analytical tools, such as convolutional neural networks (CNN) and Shapley Additive Explanations (SHAP) values for feature attribution. However, the field appears to struggle with embracing the current **large language models (LLMs) and vision-language models (VLMs)** revolution, which offers a significantly broader array of sophisticated tools and enhanced capabilities for medical AI applications. This technological gap represents both a challenge and an opportunity for advancing XAI in cardiovascular healthcare.

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Authors Lorenzo Bertolini, Valentin Comte, Victoria Ruiz-Serra, Lara Gierschmann, Lia Orfei, Sergio Consoli, Mario Ceresa, Maddalena Querci, Tobias Wiesenthal

1 Introduction

Artificial intelligence (AI) has rapidly grown in power and capability over the past decade and is now pervasive in everyday life and work environments. This technological advancement can be particularly impactful in medical and health research applications, where AI systems can increasingly be deployed to assist healthcare systems under rising pressure with diagnosis, treatment planning, drug discovery, and patient monitoring [51, 23].

However, with this increased power comes dramatically increased complexity. This increased complexity results in many modern AI models becoming "black-boxes", meaning that their internal decision-making processes are opaque and difficult to understand, even for their creators. The sophisticated neural networks and deep learning architectures that enable these remarkable capabilities operate through millions or billions of parameters and complex mathematical transformations that resist straightforward interpretation [50].

Understanding how these black-boxes work internally and obtaining relevant and meaningful explanations that describe how inputs are transformed into outputs is a critical challenge for building trust in AI models. Building more transparent and understandable explanation and interpretations of AI models might prove to be beneficial to help build trust in sensitive domains such as healthcare and medicine, where decisions can have life-or-death consequences and regulatory oversight demands transparency and accountability.

These challenges have given rise to several interconnected AI research fields, including interpretability, explainability, trustworthiness, and fairness in AI systems. Each of these areas addresses different aspects of making AI more transparent while presenting multiple overlaps and shared goals. However, despite their growing prominence in academic literature and industry discussions, significant confusion and overlap remain in how these terms are defined and applied [14, 37, 26, 5, 46, 47, 51].

The healthcare sector presents unique challenges and opportunities for these AI transparency efforts. Medical applications require not only accurate predictions but also explanations that clinicians can understand, validate, and incorporate into their clinical reasoning. Patients and healthcare providers need to trust that AI recommendations are based on sound medical principles rather than spurious correlations or biased training data [43, 12, 51].

In this context, cardiovascular disease (CVD) represents a particularly important application area. As one of the leading causes of mortality worldwide [45], CVD affects millions of patients and presents complex diagnostic and therapeutic challenges that could benefit significantly from AI assistance [46]. However, the high stakes involved in cardiovascular care make the need for explainable and trustworthy AI systems even more critical.

While this report primarily focuses on AI models and XAI methods applied to images and text in healthcare, it is important to acknowledge that other highly impactful AI models exist beyond these modalities. Notably, AI-driven biological models have recently achieved significant breakthroughs, exemplified by the 2024 Nobel Prize in Chemistry awarded for advances in protein structure prediction and design that hold great promise for accelerating biomedical research and informing future clinical applications, from personalised therapeutics to novel treatment strategies. Despite their transformative potential, the current implementation of these AI biology models in routine clinical practice remains limited. Consequently, they are not explored in detail within this report. For readers interested in the latest developments in AI for science, including the field of protein structure prediction, we refer to the JRC technical report [1].

[1] "The Role of Artificial Intelligence in Scientific Research." 2025. Science-for-policy Report, JRC143482 (Under Review).

This report aims to address the current landscape of AI models and the available interpretability and explainability tools, with a focus on healthcare and particular attention to cardiovascular applications. We seek to provide clear definitions of key concepts, examine the available methods and techniques for achieving transparent, explainable, and interpretable AI, explore their main applications to health and medical research, and investigate how these approaches can be specifically tailored to address the unique challenges and opportunities in cardiovascular disease management and research. We also propose a suitable way forward to ensure the effective deployment of these rapidly evolving technologies.

1.1 Interpretability and Explainability

As mentioned in the introduction, interpretability and explainability remain somewhat ambiguous and not yet fully defined. In the ISO 29119-1 technical report [1], explainability is defined as the *"property of an AI system to express important factors influencing the AI system results in a way that humans can understand"*, while interpretability is described as the *"level of understanding how the underlying (AI) technology works"*.

These definitions are accessible and offer valuable clarity. However, given the complexity and vastness of these fields of research, some fundamental aspects of how these two areas are currently applied and understood remain to be clarified. For instance, explainability has traditionally been associated with post-hoc analysis of an existing model using another model, with the objective of obtaining insights and explanations about how the model reached a decision or, more broadly, how the model's input and output are connected. Intuitively, by post-hoc, we refer to any process, including obtaining an explanation, that takes place after the model has already processed an input and produced an output. The post-hoc nature of current XAI approaches hardly aligns with the ISO definitions, which appear to be more intrinsic to the model itself.

Similarly, interpretability has been generally associated with understanding how the inner mechanisms of an AI model function. However, this definition is traditionally associated with transparent models. We typically define transparent or white-box models as those models for which we fully know the "meaning" and impact of each model parameter. An example would be a linear regression model, where we are fully aware of the role of the learned bias and intercept terms, and we can understand the impact that changing one input feature will have on the model's output. Since machine learning (ML) and AI systems have evolved into more complex black-boxes with large numbers of parameters, the proposed definition of interpretability has become harder to apply.

Today, a straightforward solution to this definitional conundrum is to base the distinction between these two areas—whose boundaries remain blurred—on the semantics of their root words. In essence, we can state that interpretability involves understanding, or *interpreting*, the inner processes, mechanisms, and components of a model, whether causally or correlationally. Explainability, on the other hand, focuses on understanding, or *explaining*, the decision process of a model, with or without unveiling the inner processes that led the model to a specific output. A more formal distinction would be that interpretability examines the extent to which we can establish correlational or causal effects within a system's internal processes by interpreting how its parameters change given changes in input, while explainability defines the extent to which the behaviour, internal state, and mechanisms of the model can be explained in the context of its decision process and understood in human terms.

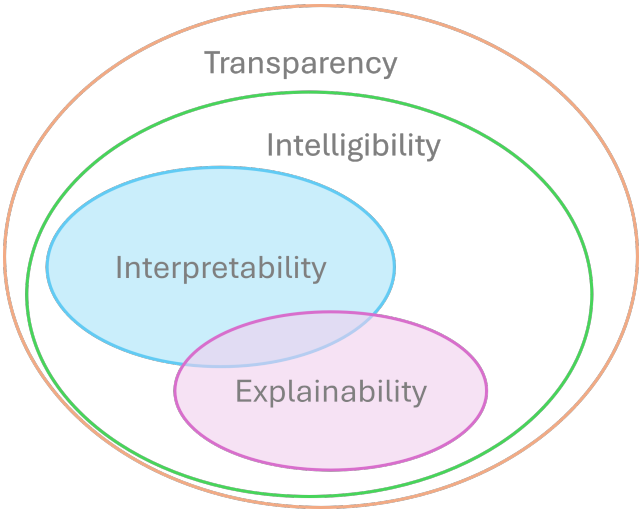
As will become clearer throughout this report, these two terms do not establish a clear-cut dichotomous distinction, but are more useful when viewed as two ends of a blurred continuum. Indeed, some of the methods presented in this work can be easily categorised under a specific framework. Take circuits [28,

57, 22, 59], for instance. They reveal which specific set of model components (called *circuits*) in large language models causally contributes to a specific behaviour or ability. However, they do not truly *explain* how this process occurs and can hence be considered an interpretability method. For other methods and tools, the distinction appears more opaque, as is the case with Shapley additive explanations (SHAP) [36], which the literature has identified and used within both explainability and interpretability frameworks. Given this definitional ambiguity, for the sake of clarity, throughout this report we will use the term XAI to encompass all methods and techniques related to both explainability and interpretability

1.2 Trustworthiness, and the Interplay with XAI

Trustworthiness and explainability are extremely interrelated concepts, both in general AI applications and in health-focused AI systems. When we consider the relationship between explainability or interpretability and trustworthiness or transparency, we intuitively envision a hierarchical structure, similar to the one illustrated in Figure 1 [14], or the one proposed by Sun and colleagues [51], from Figure 2. This conceptual framework demonstrates how explainability serves as a foundational pillar that supports the broader architecture of trust in AI systems.

Figure 1: Ven diagram describing the interplay between interpretability, explainability, and transparency in AI systems.

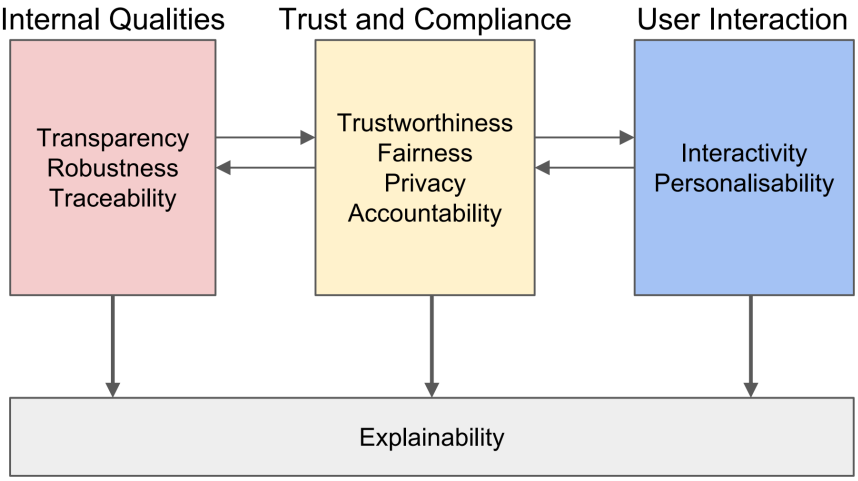


Source: adapted from Clinciu and Hastie, 2019 [14]

This relationship is particularly significant in healthcare, where AI-driven decisions carry exceptionally high stakes, and the consequences of misplaced trust can be life-altering. Efficient explanation mechanisms are essential to ensure the transparent and accountable use of AI systems by healthcare practitioners, establishing a bridge between complex algorithmic processes and clinical decision-making. When medical professionals can understand how an AI system arrives at its conclusions—whether through visualisations of decision pathways, confidence scores, or feature importance rankings—they can make more informed decisions about whether to trust and act upon those recommendations.

The importance of this understanding goes beyond mere technical comprehension. Healthcare providers must be able to critically evaluate AI outputs within the context of their clinical expertise, patient history, and situational factors that the AI system may not fully capture. This collaborative approach between human expertise and artificial intelligence creates a more robust diagnostic and treatment framework than either could achieve independently. Community bridging initiatives, for example, private-public partnerships and educational collaborations as described in a recently published JRC report [27], can support this

Figure 2: Interrelationship among XAI-related terms.



Source: Sun et al., 2025 [51]

by jointly developing skills, assessment frameworks, and guidance tailored to clinical realities and emerging AI technologies.

Furthermore, a well-designed suite of explainability and interpretability tools serves multiple purposes beyond professional accountability. These tools not only enable a more trustworthy AI-supported diagnosis but also enhance patient participation in the medical decision-making process, fostering a more collaborative healthcare environment. When patients can understand the reasoning behind AI-assisted diagnoses or treatment recommendations—presented in accessible language rather than technical jargon—they feel more involved and empowered in their own healthcare journey. This transparency can help alleviate patient concerns regarding "black-box" medical decisions and reduce anxiety about AI involvement in their care.

The benefits of this enhanced patient empowerment extend to improved therapeutic relationships between patients and providers, as shared understanding of decision-making processes builds mutual trust and confidence. Patients who comprehend the rationale behind their treatment are more likely to adhere to prescribed therapies and maintain active participation in their care plans [51], ultimately leading to better therapeutic relationships and potentially improved health outcomes. This creates a positive feedback loop where transparency fosters trust [12], trust enhances compliance, and compliance contributes to improved clinical outcomes. From a broader perspective, XAI methods and tools can contribute to enhancing transparency and fairness, which are two fundamental pillars among the seven non-binding ethical principles for artificial intelligence described in the High-Level Expert Group on Artificial Intelligence (AI HLEG), and supported through the AI Act (Regulation (EU) 2024/1689). These principles serve as a comprehensive framework designed to ensure that AI systems remain trustworthy, ethically sound [21], and aligned with human values as they become increasingly integrated into critical decision-making processes across various sectors.

2 AI Architectures and XAI Methods

This section presents an overview of the main AI models adopted by researchers and practitioners, alongside the principal categories of interpretability and explainable AI methods that have been proposed and developed in the literature. These approaches encompass several distinct methodological frameworks that collectively aim to illuminate the inner workings of complex AI systems. In summary, the models and methods discussed include feature-based methods, visual interpretability methods, and mechanistic interpretability methods.

Feature-based methods represent one key category, with techniques such as SHAP serving as prominent examples. These methods quantify the relevance and contribution of individual input features in determining the model's outputs and predictions, providing stakeholders with clear insights into which variables most significantly influence algorithmic decisions.

Visual interpretability methods constitute another important approach, exemplified by techniques like GradCAM (Gradient-weighted Class Activation Mapping) [48]. These methods enable researchers to visualise and "see" where models "pay attention" during the decision-making process, effectively opening windows into the black-box of neural network architectures. By generating heat maps and attention visualisations, these techniques reveal which regions of input data most strongly activate the model's decision pathways.

Mechanistic interpretability methods offer an additional perspective, with sparse autoencoders (SAEs) features within neural networks [50] representing a cutting-edge example of this approach. These techniques enable researchers to understand which latent representations and concepts emerge within the model's internal activations, providing granular insights at the level of individual neurons and network components. This approach is particularly valuable for understanding how models develop internal representations of complex concepts and relationships.

Together, these diverse methodological approaches provide a multifaceted framework for understanding and interpreting the decision-making processes of complex AI systems, enabling both technical practitioners and end-users to build greater trust and comprehension of AI-driven outcomes.

2.1 AI Architectures

Before the emergence of modern large language models (LLMs) [56, 18, 9], several artificial neural network architectures dominated both research and clinical practice. Among these, convolutional neural networks (CNNs) [35] have emerged as a predominant architecture for addressing various medical and healthcare challenges, primarily due to their demonstrated effectiveness in processing and analysing complex medical imagery [16]. These CNN-based approaches established themselves as the gold standard across a wide spectrum of healthcare applications, including diagnostic imaging, pathology detection, radiological analysis, and computer-aided diagnosis systems.

In recent years, there has been a significant paradigm shift toward transformer-based models [56] across AI applications generally, with particularly notable adoption in the healthcare domain. This transition has been driven by transformers' remarkable versatility and their superior ability to process diverse inputs in a unified, multimodal framework. Unlike their predecessors, these architectural innovations have enabled the development of increasingly sophisticated systems capable of simultaneously handling heterogeneous data types. These systems can seamlessly integrate textual information, medical images, time-series physiological data, electronic health records, and clinical notes within a single coherent framework.

The following sections will provide an overview of four principal AI model architectures that currently dominate the literature and find widespread implementation in contemporary healthcare applications.

We will examine their fundamental design principles, core computational mechanisms, and specific advantages in medical contexts [40, 32, 30, 13].

Developing a high-level yet thorough understanding of how these models are constructed and operate is essential for comprehending the functionality and effectiveness of various XAI tools, as well as the types of insights they can provide to clinicians and researchers. This foundational knowledge is crucial because many XAI techniques are more effective when applied to specific model components, while others were explicitly designed to interpret and elucidate particular architectural elements of these complex systems. Furthermore, the choice of XAI method often depends on the underlying model architecture, making this background knowledge indispensable for selecting appropriate interpretability approaches in healthcare AI applications.

2.1.1 Convolutional neural network

Convolutional neural networks (CNN) [35] are among the most adopted artificial neural network architectures, and have become the foundation of AI applications in medical imaging, thanks to their ability to automatically learn complex patterns and features from detailed visual data [16]. Unlike traditional image processing methods that depend on manually designed filters, CNNs learn these representations directly from data through an end-to-end training process. This capability is particularly valuable in medical settings, where subtle variations in tissue texture, shape, or contrast can reveal the presence of disease.

CNNs work in a way that resembles the human visual system. Early layers focus on basic features such as edges or textures, while deeper layers gradually combine these into more complex representations like shapes or anatomical structures. This step-by-step processing allows CNNs to capture spatial patterns at multiple levels, which is crucial when analysing detailed medical images such as MRI, CT scans, or X-rays.

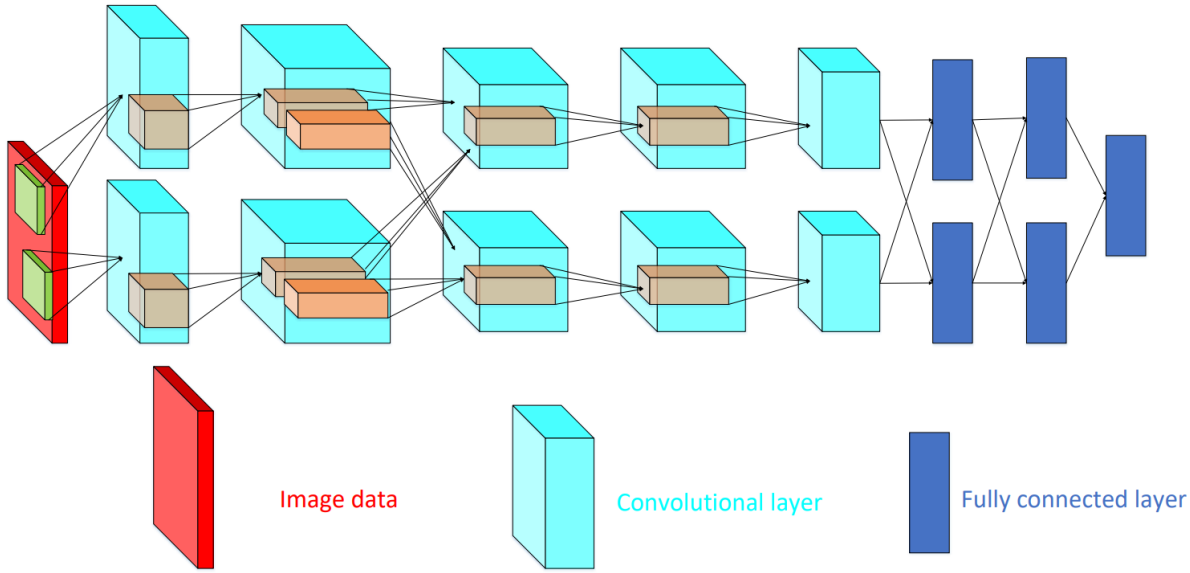
A CNN is built from repeated building blocks, each containing a convolutional layer, an activation function, and a pooling layer. The convolutional layer uses small, trainable filters that move across the input image to create feature maps, capturing important local patterns such as edges or textures. Each unit in the convolutional layer processes information from a specific region of the image, called the receptive field, allowing the network to learn spatially focused features effectively.

Figure 3 shows the complete architecture of AlexNet [53], a CNN for image classification, demonstrating how input images are processed through multiple layers of convolutions and pooling before reaching fully connected layers that produce the final prediction. This architecture is now widely used and adapted for various medical imaging tasks such as tumour classification, organ segmentation, abnormality detection, and image registration [16]. Together, these capabilities make CNNs highly effective for automating and improving clinical workflows. By detecting subtle patterns that may not be visible to the human eye, CNNs contribute to more accurate diagnoses and support clinical decision-making across a wide range of medical applications.

2.1.2 Transformer and attention mechanism

The transformer architecture was first introduced by Vaswani et al., in their seminal work “Attention is All You Need” [56]. The complete architecture, summarised in Figure 4, consists of two main components: an encoder and a decoder block, both constructed from N transformer layers. While the encoder and decoder blocks differ in their specific implementations, the transformer layers within each block are built

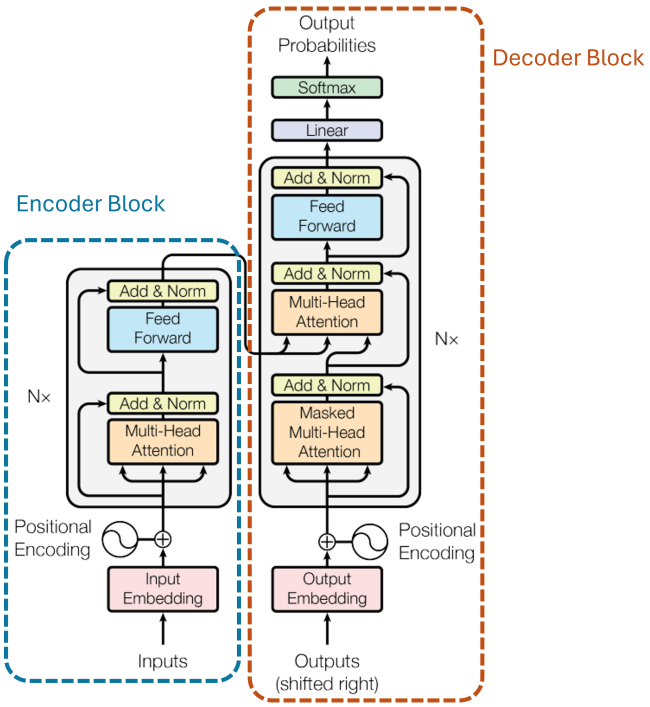
Figure 3: Diagram describing the implementation of the AlexNet CNN.



Source: Tang et al., 2023 [53]

from the same two fundamental components: a simple feed-forward neural network and the multi-head self-attention mechanism.

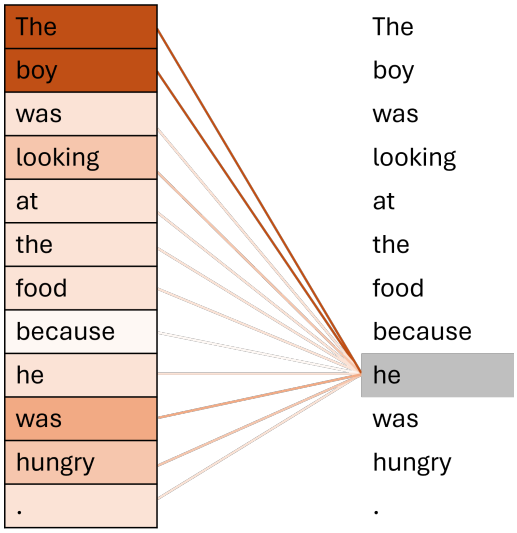
Figure 4: Diagram describing the first implementation of a transformer neural network in its sequence-to-sequence form.



Source: adapted from Vaswani et al., 2017 [56]

Intuitively, given an input sentence, the self-attention mechanism develops computational representations for each word in the sentence by learning from its surrounding context where to focus attention, as illustrated in the diagram in Figure 5. For instance, consider the sentence "`The boy was looking at the food because he was hungry.`". In the example presented in Figure 5, we observe how the algorithm, when encoding the representation for the word "`he`" at a given layer n, pays greater attention to (or draws more information from) words such as "`The`" and "`boy`", which provide contextual clues about the pronoun's referent.

Figure 5: Diagram describing the self-attention mechanism. The figure shows how, to encode the representation for the word "`he`" at the decoder layer *n*, the algorithm pays more attention (or uses more information from) words like "`The`", "`boy`".



Source: own elaboration

The diagram in Figure 5 also illuminates one of the key distinctions between encoder and decoder blocks. In the example, we see that despite focusing primarily on words preceding "`he`", the model allocates some attention to words that follow "`he`". This bidirectional attention pattern is characteristic of encoder blocks, which can access the entire input sequence simultaneously. Decoder models, conversely, implement a causal attention mechanism that allows a word to draw information only from words that preceded it in the sequence, ensuring that the generation process respects temporal dependencies.

Throughout the years, researchers have developed transformer-based LLMs that utilise different architectural configurations. These include encoder-only models like BERT [18], which excel at understanding and encoding text representations; encoder-decoder models (also known as sequence-to-sequence or seq2seq models) like T5 [41], which can both understand and generate text; and decoder-only models like GPT [9], also referred to as generative or autoregressive models.

It is crucial to understand that among these three types of models, only seq2seq and autoregressive models are considered generative AI (GenAI) models, as they possess the capability to generate novel text, images, or other content. Encoder models, by contrast, are designed to encode data from a specific modality into meaningful representations and are typically employed for tasks such as sentiment analysis, text classification, or recommendation systems [56, 18, 41]. To create computational representations of long sequences of text (embeddings), encoder models typically prepend a special token called the `CLS` (classification) token to input sentences. This token does not represent any specific word in the input sentence but serves as a global aggregator that captures information from the entire sequence, producing a

compressed and encoded representation of the overall input. In AI and NLP, a token is the smallest meaningful unit into which text is broken down for processing by machine learning models. In the context of this working example, the concept of tokens overlaps with that of words or punctuation, as shown in Figure XXX. However, the concept is slightly more complex in practice. To solve the problem of unseen new words, LLMs are trained to split words in ways that are not always morphologically sound (e.g., “playing” to “play”, “ing”), and instead opt for statistically useful splitting (e.g., “playing” to “play”, “i”, “ng”).

It is important to note that the diagram in Figure 5 presents a schematic representation of a single attention head. However, the multi-head attention mechanism typically comprises multiple attention heads operating simultaneously, each attending to different aspects of the input sequence and drawing complementary information. This parallel processing allows the model to extract diverse types of patterns and relationships occurring throughout the input sentence. Collectively, these attention heads distributed across the model enable different layers of the transformer to specialise in specific linguistic aspects, such as syntax, grammar, semantics, or discourse structure [55].

Understanding the principles underlying the attention mechanism and the broader transformer architecture is particularly important because a significant proportion of XAI tools designed to unveil the internal behaviour of transformer-based LLMs focus specifically on interpreting and visualising attention patterns. These interpretability methods leverage our understanding of attention mechanisms to provide insights into how these models process language and make decisions, making the attention mechanism a critical component for both model performance and explainability.

2.1.3 Vision transformer

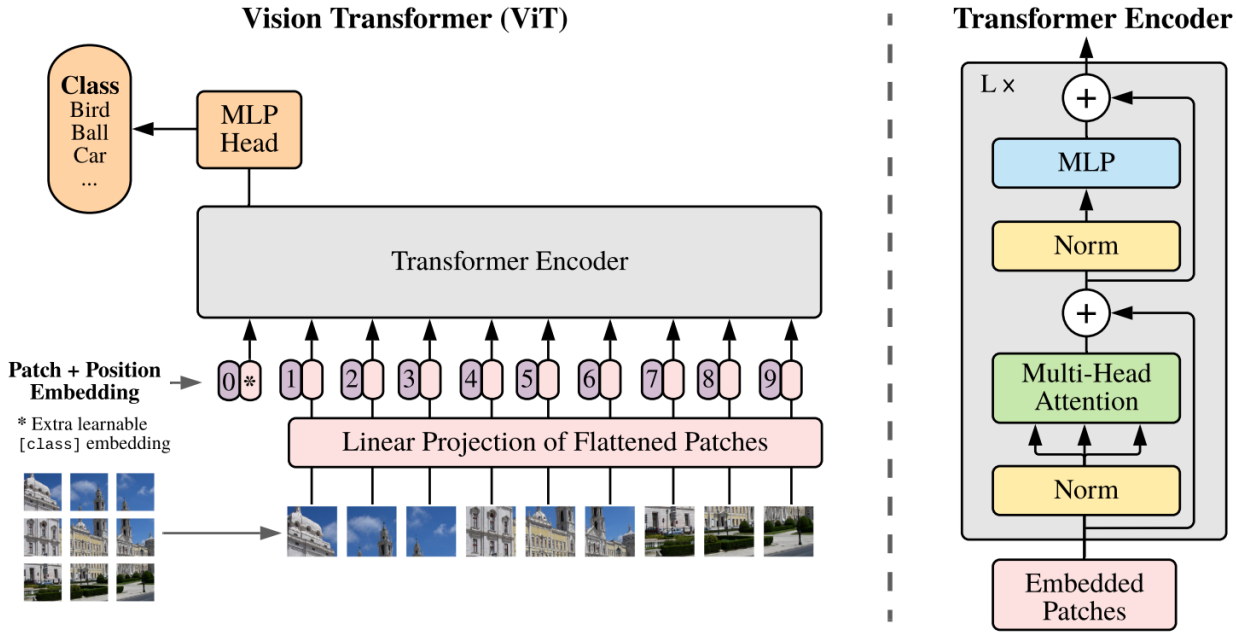
Encoder transformer models demonstrate exceptional effectiveness for classification tasks [18, 41], which represent a fundamental application area in medical AI systems. Moreover, as we have observed, encoder models possess the unique capability to encode information through comprehensive reasoning across the entire context of an input, simultaneously extracting and integrating information from all parts of the input sequence. This global contextual understanding was one of the primary limitations identified in CNNs, which typically process information through localised filters and struggle to capture long-range dependencies and global patterns within the data.

This limitation was a driving factor that motivated researchers to adapt encoder-transformer architectures, originally designed for natural language processing tasks, to handle visual data such as medical images. The pioneering implementation of these adapted models was the Vision Transformer (ViT), whose architecture is summarised in Figure 6.

The overall ViT architecture shares substantial similarities with the encoder blocks illustrated in Figure 4. However, a crucial distinction lies in the initial processing stage of the algorithm. Unlike standard text transformers that work directly with token embeddings, ViT incorporates an additional preprocessing step called the Linear Projection layer. This layer serves as an interface between visual and textual modalities by taking image patches as input and converting them into a format compatible with transformer processing. Specifically, the Linear Projection layer flattens image patches and projects them into a fixed-dimensional embedding space, essentially treating each image patch as a “visual token” that can be processed by the subsequent transformer layers.

The ability of ViT and other transformer-based vision models to simultaneously reason over the whole (visual) input has made these new artificial neural networks incredibly popular in image-based tasks. Their

Figure 6: Diagram describing the first implementation of a vision transformer neural network.



Source: Dosovitskiy et al., 2021 [19]

application has significantly improved the processing of medical images, in fields like cancer and cardiovascular disease detection, as recently shown by a report published by the Joint Research Centre (JRC) [16].

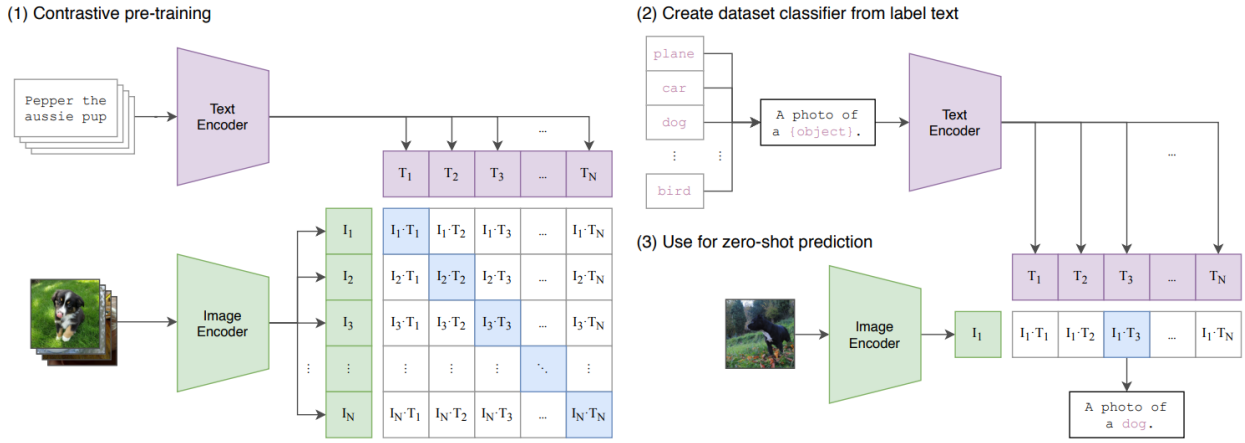
2.1.4 CLIP and vision-language models

Many general-purpose or health-related tasks require analysing data that combines both modalities. For instance, in disease classification tasks, models would benefit from incorporating not only metadata such as gender, ethnicity, or age, but also visual symptoms and clinical descriptions. However, despite being based on slightly different variations of the same neural network architecture, encoder-only LLMs and ViT models can only process one modality at a time—either textual or visual data.

A computationally efficient solution that avoids the overhead of heavy generative models is to employ dual-encoder vision-language models (VLMs), such as the Contrastive Language-Image Pretraining model (CLIP) [40]. CLIP models utilise a learning method that teaches a model to find the correct pairs between two sets of items from different data modalities, such as a list of dog images and a list of textual descriptions of dog breeds. This method simultaneously trains separate textual and visual transformer encoders to produce unified multimodal representations from unimodal inputs. The contrastive learning process works by presenting the model with sets of paired textual descriptions and images, training it to identify which text-image combinations belong together while distinguishing them from mismatched pairs (see Figure 7.1).

Once trained, CLIP models function as versatile classification systems through zero-shot prediction—the ability to classify new categories without additional training on specific tasks. This is accomplished by encoding both the input data and potential class labels into the same representation space, then measuring

Figure 7: Diagram describing the first implementation of a vision CLIP neural network.



Source: Radford et al., 2021 [40]

their similarity (see Figures 7.2-7.3). Despite their relatively compact size compared to large generative models, CLIP architectures demonstrate remarkable power and versatility, leading to widespread adoption in healthcare and medical applications [32, 30], including cardiovascular diagnostics [13].

2.2 Feature-based XAI Methods

Feature-based attribution methods represent a foundational category of XAI techniques that systematically quantify how individual input features contribute to model predictions [44, 36, 4]. These methods function as crucial intermediaries between the opaque decision-making processes of complex models and the interpretability requirements of human users, achieving this by assigning importance scores or relevance values to distinct components of the input data.

The concept of "features" within this framework operates across multiple levels of abstraction, providing remarkable flexibility in analysis approaches. At the most granular level, features correspond directly to the raw input variables present in datasets, such as a patient's age in medical records, individual pixel intensities in digital images, or term frequencies in textual corpora. However, the scope of feature-based attribution extends far beyond these elementary representations to encompass more sophisticated, semantically rich constructs that are dynamically extracted or constructed by the explainability framework itself.

A particularly compelling illustration of this semantic sophistication can be found in Concept Activation Vectors (CAVs) [33], which have proven especially effective for interpreting deep neural networks in image classification applications. CAVs illustrate how feature-based methods can go beyond simple input-level analysis to identify and quantify higher-order conceptual patterns, aligning more closely with human cognitive frameworks.

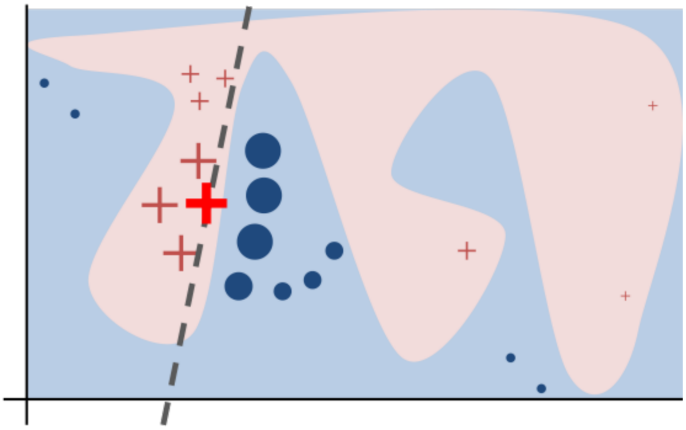
This inherent flexibility in operating across different levels of abstraction makes feature-based attribution methods exceptionally valuable for two purposes: elucidating the mechanical operations of model architectures and ensuring alignment with human conceptual understanding. This bidirectional interpretability capability is essential for establishing trust in AI systems, especially in high-stakes domains where decisions must be transparently communicated to domain experts, regulatory bodies, and end users. By bridging the gap between computational complexity and human comprehension, these methods serve as critical enablers for the responsible deployment of AI technologies across diverse application domains.

2.2.1 Local interpretable model-agnostic explanations (LIME)

Local Interpretable Model-Agnostic Explanations, or LIME [44], is an explainability tool designed to quantify the impact that specific features of any black-box ML model have on the model's output. As a post-hoc method, LIME does not require access to the internal structure or parameters of the model that generated the prediction—only access to its outputs. This model-agnostic approach makes LIME particularly valuable for interpreting complex models where internal mechanisms are opaque or proprietary.

The fundamental steps and procedures underlying LIME's methodology are both simple and intuitive. Given a real instance from the dataset, LIME begins by generating a sample of artificial instances through systematic perturbation of the original data point. These perturbations modify feature values while preserving the overall structure of the input space. LIME then observes how the black-box model responds to these artificial examples, collecting the corresponding predictions. Using this data, LIME fits a linear model—which is inherently interpretable—to approximate and explain the black-box model's behaviour in the local neighbourhood around the instance of interest.

Figure 8: Diagram showcasing the intuition behind LIME classification procedure. Red crosses and blue dots represent instances belonging to the two classes. Similarly, the red-blue shaded area represents the original classification plane produced by the model we want to interpret. To simplify the understanding of the classification learned by the model, LIME focuses on a specific part of the space—the one around larder icons—where the separation between the two classes can be simplified, and represented with a linear approximation—the dotted line.

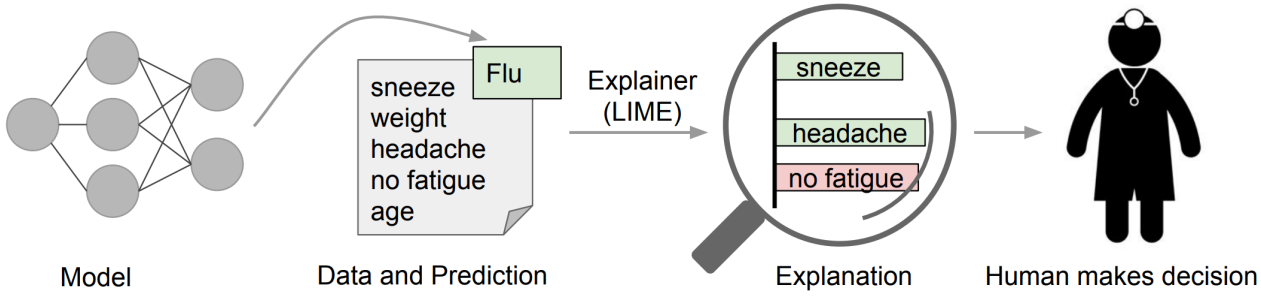


Source: Ribeiro et al., 2016 [44]

The key intuition behind this system is straightforward and elegantly illustrated in Figure 8. While the global decision boundary of a complex model (represented by the shaded sections) may be intricate and non-linear, focusing on a localised region (such as where crosses and dots appear larger in the figure) often reveals that the classification behaviour can be effectively approximated by a simple linear boundary. This local linear approximation serves as the foundation for LIME's explanations.

A linear model learns coefficients that directly indicate the weights associated with each input feature, providing a clear mathematical interpretation of feature importance. Figure 9 demonstrates how LIME's workflow operates in practice. Since LIME relies on linear approximations, the system provides two primary types of information: feature relevance (indicating how important each feature is for supporting a particular class, represented by the height of rectangles in Figure 9) and feature direction (indicating which class each feature supports, represented by the colour of rectangles in the same figure).

Figure 9: LIME workflow and usage example.



Source:Ribeiro et al., 2016 [44]

However, LIME has an important limitation: it does not quantify *how much* each feature has *contributed* to the actual prediction output in absolute terms. Instead, it provides relative importance scores within the local linear approximation. This distinction is crucial because LIME's explanations are inherently local and approximate, focusing on feature relevance rather than precise contribution measurement. For more precise attribution information that quantifies actual feature contributions to predictions, alternative explainability tools such as SHAP.

2.2.2 Shapley Additive Exlanations (SHAP)

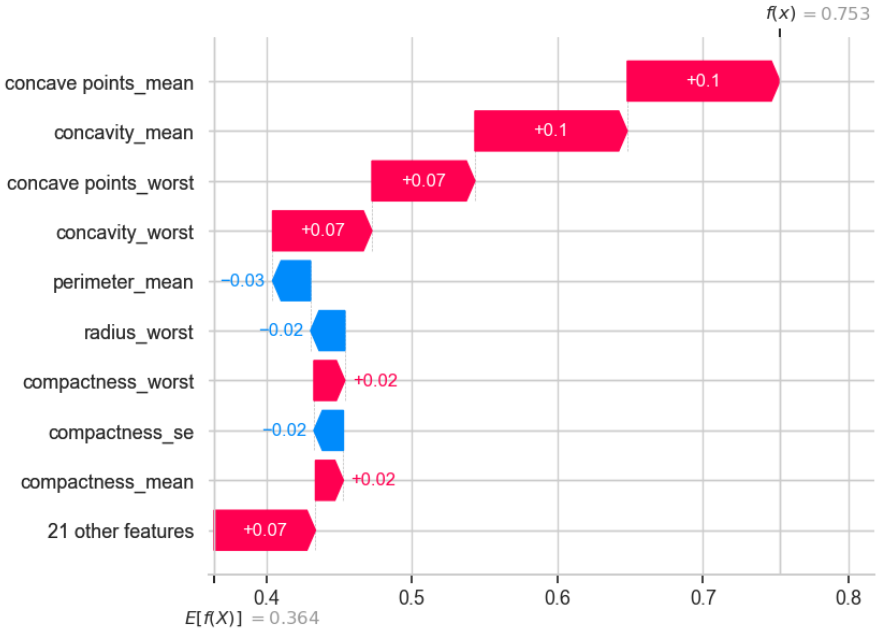
Similar to LIME, Shapley additive explanations (SHAP) [36] is an interpretability tool based on Shapley values [49] that aims to estimate the impact, or *contribution*, of specific features on final predictions across all possible combinations of features. Intuitively, SHAP quantifies the effect of removing a particular feature from the input on a prediction model's output. Unlike simpler feature importance methods, SHAP not only estimates the individual impact of each feature but also how feature interactions contribute to the model's decision-making process.

This estimation is performed by computing the marginal contribution of a feature across different subsets of the feature space. Specifically, SHAP calculates the difference in model predictions when a feature is present versus absent, considering both the complete feature set and various subsets that exclude the feature of interest. For example, consider a system trained to predict cardiac arrest risk using 30 features, including patient age. To estimate age's contribution for a specific patient, SHAP would compute age's impact when combined with all other 29 features, as well as when paired with smaller subsets of 15, 5, or 2 features. This evaluation across different feature combinations provides a robust measure of each feature's true contribution to the prediction.

Compared to LIME, SHAP offers several distinct advantages. Most notably, beyond providing feature importance rankings, SHAP quantifies the actual magnitude and direction of change that each feature contributes to the final prediction (see Figure 10). Additionally, SHAP provides both local explanations (for individual instances) and global explanations (aggregated insights across the entire dataset or task), offering a more comprehensive interpretability framework. The method does not require training separate surrogate models, unlike LIME's approach, and includes multiple model-specific algorithms optimised for different ML architectures, such as TreeSHAP for tree-based models and DeepSHAP for neural networks.

However, despite these algorithmic optimisations, computing exact SHAP values remains computationally expensive due to the exponential complexity inherent in evaluating all possible feature combinations. This

Figure 10: Shape single-instance output. The graph shows how much each feature (Y-axis) contributes positively (pink rectangle) or negatively (blue rectangle) to the final predicted score for a single instance x (f(x)=0.753, upper right).



Source: https://medium.com/@naveed88375/deep-learning-model-interpretability-with-shap-63598b7aeff8

computational burden can become prohibitive for high-dimensional datasets or when real-time explanations are required, representing the primary limitation of the approach.

2.2.3 Concept extractions

The features analysed by XAI methods thus far have been fully explicit. That is, they constituted direct components of the input to the AI model. However, several approaches have attempted to extract and interpret features that influence the decision-making process from *within* the model itself. These methods are primarily employed to interpret image-based inputs by utilising semantic concepts or word-grounded features to generate explanations that bridge the gap between low-level visual features and high-level human-understandable concepts.

Kim and colleagues [33] developed Concept Activation Vectors (CAVs), a technique designed to elucidate how image classification models make decisions by establishing connections between post-hoc user-defined concepts (e.g., "`striped`") and the internal representations of neural networks (e.g., when classifying an image as "`zebra`"). Their methodology involves training a linear classifier to distinguish between images containing a specific concept and those that do not, thereby revealing how these semantic concepts influence the model's final predictions. A key limitation of this approach is that the relationship between specific instances and concepts must be at least partially known a priori, preventing fully unsupervised concept extraction.

Building upon this foundational work, Yan et al., [58] developed an interactive diagnostic tool specifically tailored for skin cancer detection systems. Their framework was designed to amplify beneficial decision-making patterns while mitigating misleading visual cues that could result in misdiagnoses. The system

could automatically identify concept patterns (such as “darker border” characteristics) within different diagnostic categories (e.g., “benign” versus “malignant” lesions), providing clinicians with interpretable insights into the model’s reasoning process. However, since their approach relied exclusively on traditional convolutional vision architectures, it still necessitated domain experts to manually define and validate concepts after examining the model’s internal feature representations and learned filters.

Recently, two works have proposed similar solutions to expand CAVs to VLMs (see section 2.1.3), aiming to leverage the multimodal capabilities of these models to ground concept feature definitions in natural language. These features are subsequently used to discriminate or classify between different image categories.

Following a similar approach to CAV, Menon and colleagues [38] developed a system based on CLIP to classify individual images (e.g., ‘‘`tiger`’’, ‘‘`beach`’’), utilising complex word-based features associated with each concept (e.g., ‘‘`stripes`’’, ‘‘`has claws`’’, ‘‘`big cat`’’ for tigers, or ‘‘`waves`’’, ‘‘`shells`’’ for beaches). Rather than relying on pre-compiled datasets that pair labels with relevant concepts, the authors employed a GPT model to automatically extract class-feature pairs (e.g., {‘‘`tiger`’’: ‘‘`stripes`’’, ‘‘`has claws`’’, ‘‘`big cat`’’}). This automated approach eliminates the need for manual curation of concept-feature associations.

Subsequently, a CLIP model was utilised to encode both the images representing different classes and their corresponding descriptive features into a shared embedding space. The method then applies a simple additive decomposition technique, combined with a scoring function, to evaluate whether a given textual description is semantically connected to a specific image class. This scoring mechanism enables the system to quantify the relevance of particular concept features for image classification tasks. The complete procedure and its output are illustrated in Figure 11.

Figure 11: Multimodal automatic concept local-explanation from Menon et al., (2023).



Source: adapted from Menon and Vondrick, 2023 [38]

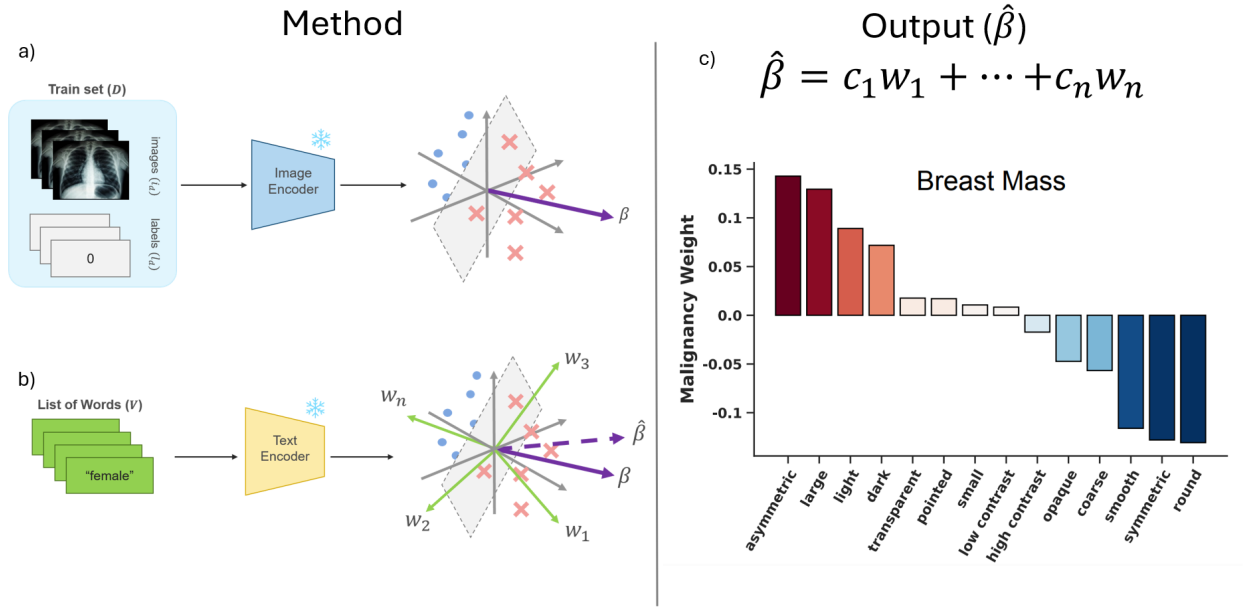
Agarwal et al., (2023) [4] proposed an innovative solution to estimate the relevance of word-grounded features in image classification tasks by leveraging the multimodal capabilities of VLMs. Similar to the approach presented in Meon et al., (2023) [38], this method employs ChatGPT to automatically generate useful word descriptions for visual features. However, to maintain maximum task agnosticism, the authors queried ChatGPT for a carefully curated set of dichotomous adjectives, such as “large” versus “small”, that describe fundamental visual properties like size, shape, and texture.

The decision to utilise such broad descriptive features stems from a key methodological distinction: unlike in Menon et al., (2023) [38], the solution proposed by Agarwal and colleagues [4] implements a *task-level*

XAI approach, a methodology that we have previously encountered only in SHAP-based implementations. To clarify this important concept, task-level explainability aims to identify which features are most relevant to the decision-making process from a holistic task perspective, rather than focusing on individual instance explanations. While SHAP achieves this by aggregating scores obtained from multiple individual instances, Agarwal's approach takes a more direct route to task-level understanding.

The methodology consists of three main steps, which are illustrated in the diagrams of Figure 12. Consider, for example, a breast cancer detection task as a concrete application scenario.

Figure 12: Agarwal et al., (2023) [4]'s method for task-level explainability is composed of three main steps. Step 1 fits the vision-based classifier and logistic regression (β). Step 2 encodes the word embeddings representing the feature, and linear modelling of those features. Lastly, step 3 is the interpretation of the linear model's coefficients (Output).



Source: own elaboration and adapted from Agarwal et al., 2023 [4]

In the first step, the authors utilise an annotated dataset where images are encoded using the image encoder component of a VLM such as CLIP. These encoded representations are then used to train a logistic classifier (β) that learns to distinguish between benign and malignant masses (Figure 12a). This classifier serves as the primary model whose decision-making process will be explained.

The second step involves leveraging the text encoder of the same VLM to obtain word representations of target concept features that were automatically generated by ChatGPT. These word embeddings are subsequently used to fit a linear model ($\widehat{\beta}$) that aims to recreate the behaviour of the original logistic classifier (β) used for medical image classification (Figure 12b). The underlying intuition is compelling: by learning a simplified linear model using interpretable words or concepts that can simulate the complex image classification model, researchers can identify which linguistic concepts are most discriminative *overall* between malignant and benign breast masses.

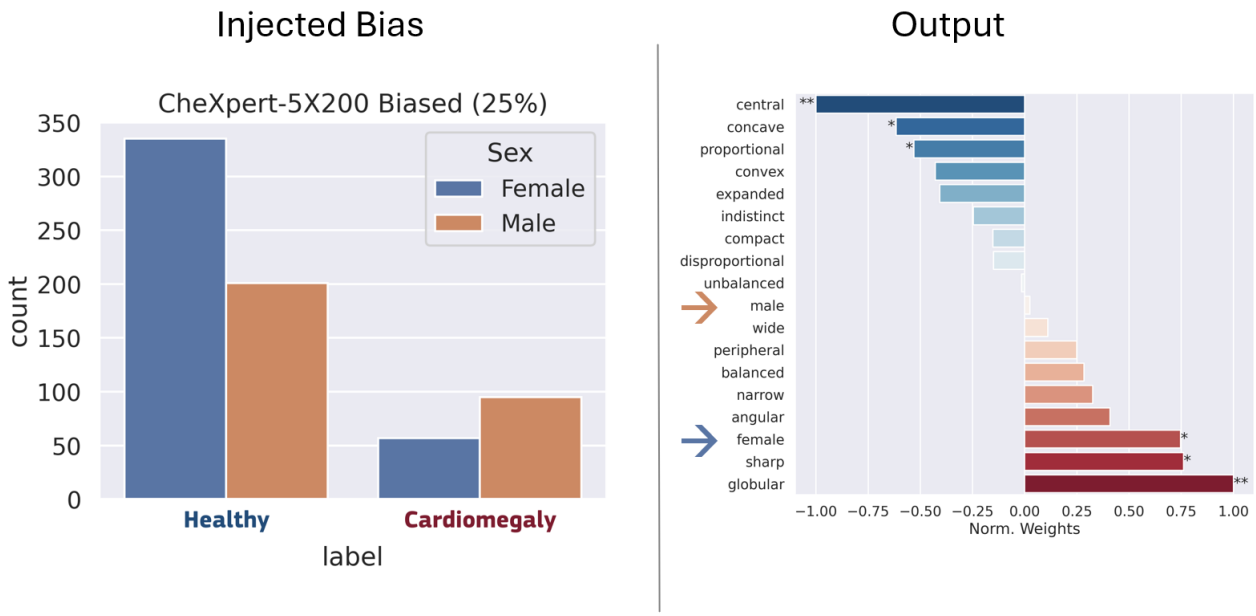
The third and final step focuses on interpreting the coefficients of the fitted linear model (Figure 12c). Since both models produce linear outputs, the results of this procedure bear resemblance to those gen-

erated by LIME, providing insights into the strength and direction of each feature's contribution toward one class or another. However, there is a crucial distinction in scope and application: while LIME enables us to understand the decision factors for a single instance from our dataset, Agarwal's method reveals the characteristic features that define each *class* across the entire task domain.

In their original work, the authors demonstrated the effectiveness of this approach across two challenging medical image classification domains: skin cancer detection and breast cancer diagnosis. The experimental results presented in the paper indicate that the system successfully identifies coherent and clinically meaningful associations, such as the relationship between malignant skin cancer and asymmetric lesion shapes. Importantly, the method also proves valuable for bias detection, successfully identifying problematic dataset artifacts such as the overrepresentation of measurement rulers in images containing malignant skin lesions—a finding that highlights potential confounding factors that could compromise model generalisation in clinical settings.

In a recent work [8], we have tested the ability of Agarwal's method to identify different types of biases injected into an X-ray dataset, focusing on a specific CVD, namely cardiomegaly. We first found that, when paired with a medical VLM, the system could correctly identify biases associated with rather explicit concepts, such as "brightness" or medical terms like "pleural effusion" and "cardiomegaly". Subsequently, we examined more implicit and potentially harmful biases connected to demographic features. To do so, we artificially increased the number of biologically male participants with the CVD while increasing the number of healthy, biologically female participants, as suggested in Figure 13 (left side).

Figure 13: Manipulated dataset (left) and results (right) from our experiment using Agarwal's method to detect biological sex biases injected in a CVD medical image task. Contrary to the injected bias, strengthening the male-cdv, and female-healthy associations, the methods associated the word "male" with healthiness, and "female" with CVD.



Source: own elaboration

Given this manipulation, we expected the method to strongly associate the word "male" with the cardiomegaly CVD (red) and similarly find a strong connection between the word "female" and the healthy

condition (blue). As shown on the right side of Figure 13, we found the opposite pattern, with “male” being more associated with healthiness and “female” closer to cardiomegaly. These results indicate that the method can be successfully used in the medical and CVD domain to discover more explicitly encoded biases, but might require more attention when investigating more sensitive biases.

2.3 Saliency-map XAI Methods

The primary objective of feature-based methods can be summarised as extracting and assigning relevance scores to components within a feature set. Indeed, we have observed how certain feature-based methods have been successfully employed to explain and interpret visual models through textual word descriptions that correspond to visual elements. While these approaches prove particularly valuable in task-level scenarios—where the goal is to achieve general discrimination between different classes or categories—they may have limitations when applied to single-instance explainability or when the attributes that characterise decisions are difficult to efficiently ground in language, such as in nodule detection or malignancy assessment.

A spatial understanding of where the model’s attention is focused can provide clearer and more intuitive insights into the decision-making process, allowing practitioners to better comprehend not only what the model predicted but also which regions or features precisely influenced that particular classification outcome.

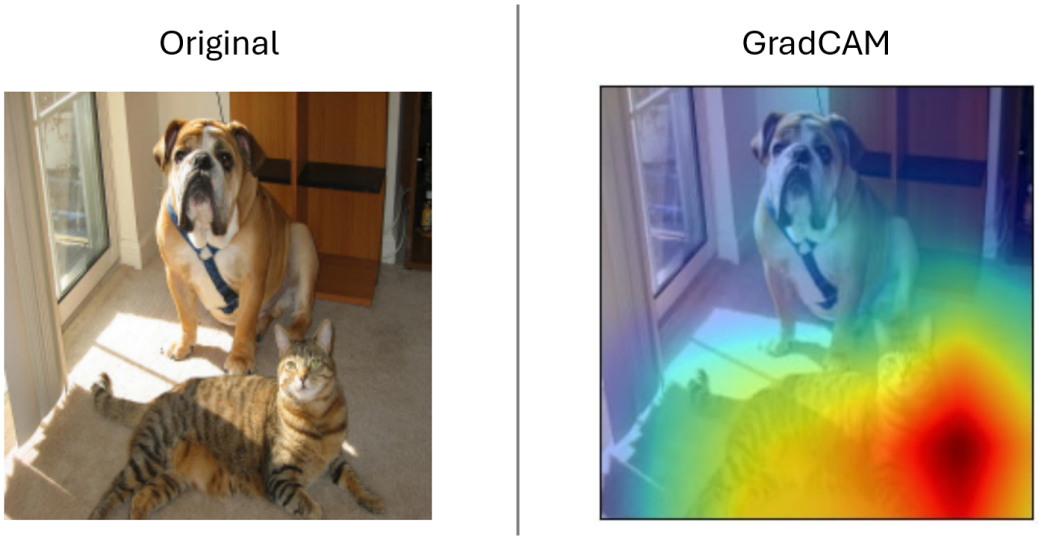
2.3.1 GradCAM

Gradient-weighted Class Activation Mapping (GradCAM) [48] is an explainability technique designed to identify which regions of an input image are considered most relevant by a neural network during the classification process. As a CNN-specific visualisation tool, GradCAM generalises previously proposed Class Activation Mapping (CAM) methods by incorporating additional gradient information. While earlier CAM algorithms typically relied solely on weights from the final convolutional layer of a CNN, GradCAM enhances these approaches by leveraging both the final layer weights and their corresponding gradients. This combination provides a more nuanced understanding of the model’s decision-making process.

Similar to other CAM-based approaches, GradCAM generates a saliency map (also known as a heat map) that visually highlights the regions of the input image that contribute most significantly to the network’s prediction for a specific class. The resulting visualisation allows researchers and practitioners to interpret which spatial features the CNN prioritises when making classification decisions. In the example shown in Figure 14, we can observe the specific regions where the CNN focuses its attention when processing gradients for the class `cat`, demonstrating the model’s learned feature representation for feline characteristics.

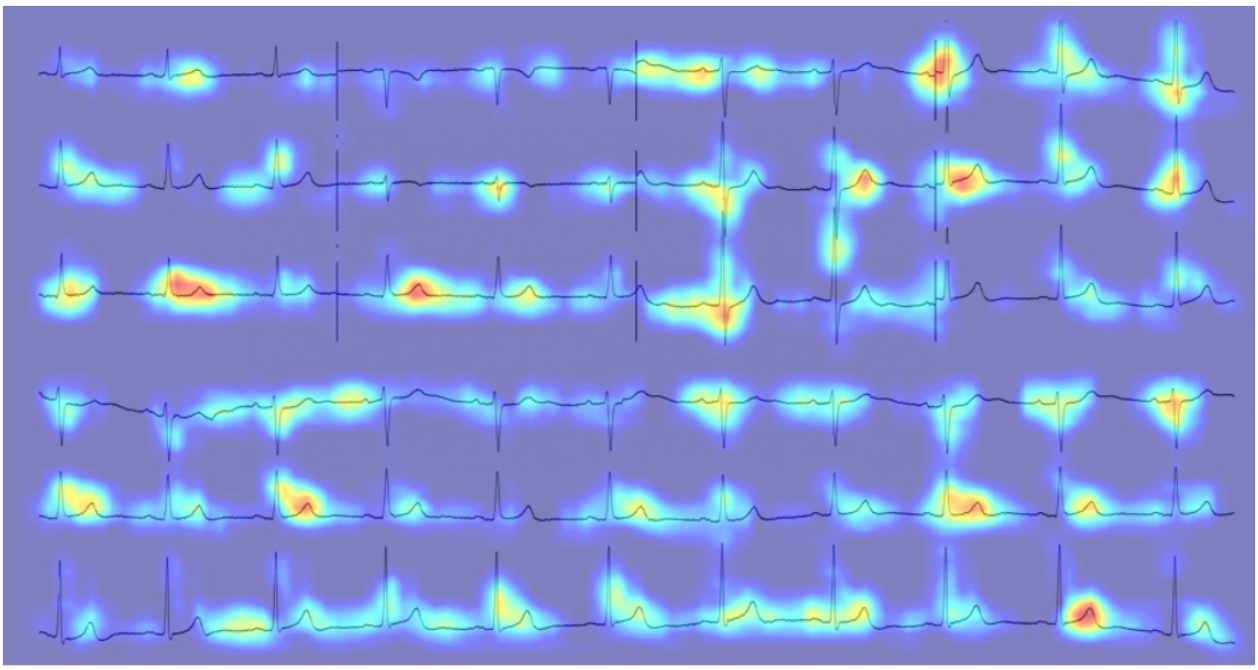
Given the widespread adoption of CNNs in healthcare and medical applications, GradCAM has become an invaluable tool for medical AI research and clinical decision support systems [46, 51]. Its ability to provide interpretable explanations for CNN predictions is particularly crucial in medical contexts, where understanding the rationale behind automated diagnoses can significantly impact patient care and clinical trust. In the medical domain, its application has been proven effective also outside its primary and most obvious data modality, images. Multiple works have, in fact, shown how it can be applied to time series data, like an electrocardiogram (ECG), a key analysis and data for cardiac investigation [46]. An example of how to use GradCAM to spot salient instances of an ECG signal can be found in Figure 15.

Figure 14: Attention map generated from GradCAM for the class `cat`.



Source: adapted from Selvaraju et al., 2017 [48]

Figure 15: GradCAM analysis for an electrocardiogram (ECG) input.



Source: Yue and Zhu, 2023 [60]

Despite GradCAM's effectiveness as a robust generalisation of earlier CAM methods and its applicability to virtually any CNN architecture, it faces limitations when applied to transformer-based models. This constraint is becoming increasingly relevant as transformer architectures gain prominence in the medical

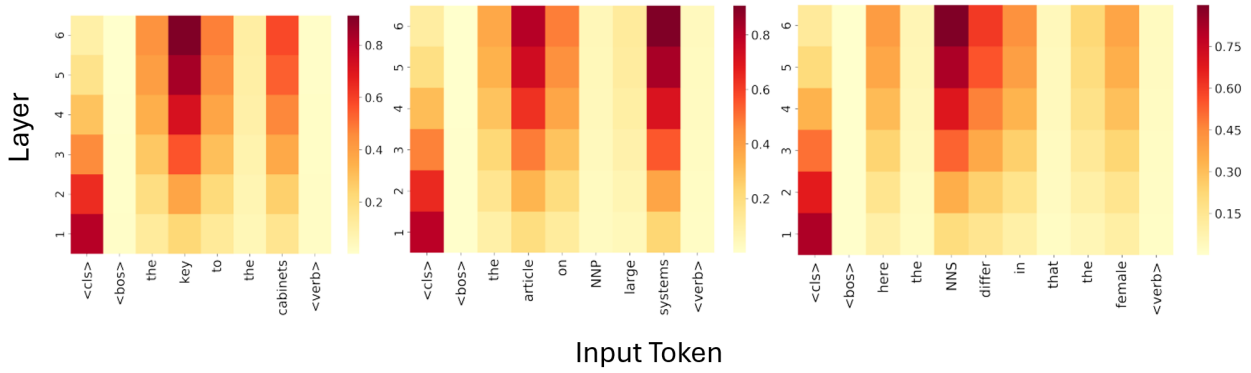
field, particularly due to their superior multimodal capabilities that can integrate diverse data types such as images, text, and clinical records for comprehensive patient assessment.

2.3.2 Attention maps

As we have seen in Section 2.1, an increasing number of works and applications in the general medical AI domain are based on transformer-based models. Thus far, the only XAI method that actively examines the model's internal activations to build explanations about the model's inference process and decision-making mechanism has been based on CNNs. In their pioneering work, Abnar and Zuidema [3] were among the first authors to suggest that the attention mechanism could be used to interpret the inner activations of transformer models and understand how knowledge flows across layers. Section 2.1.2 provides a brief, high-level introduction to the concepts underlying the multi-head self-attention mechanism. In essence, this mechanism employs multiple attention heads to modify the computational representation of each word in an input sentence across multiple layers by extracting information from (or *attending to*) the surrounding context of that word, including itself.

The fundamental intuition behind attention rollout and attention flow—the two algorithms proposed by Abnar and Zuidema [3]—is to quantify how information (or attention) flows across layers by performing a series of simple mathematical operations that weight the attention heads of layer n with those of layer $n-1$. The output of both methods consists of heat maps, one for each input word, showing how *attentive* the token in question is to every other token across all layers of the model. An example of attention rollout can be found in Figure 16. Both algorithms can be further extended to collapse information across layers, yielding a single attention-based saliency map similar to GradCAM output. However, since the input is textual in this case, the resulting map would be one-dimensional, representing the sequential nature of language data.

Figure 16: Attention rollout output for a six-layer BERT small model, and three different input sentences, each seen for the `CLS` Token.

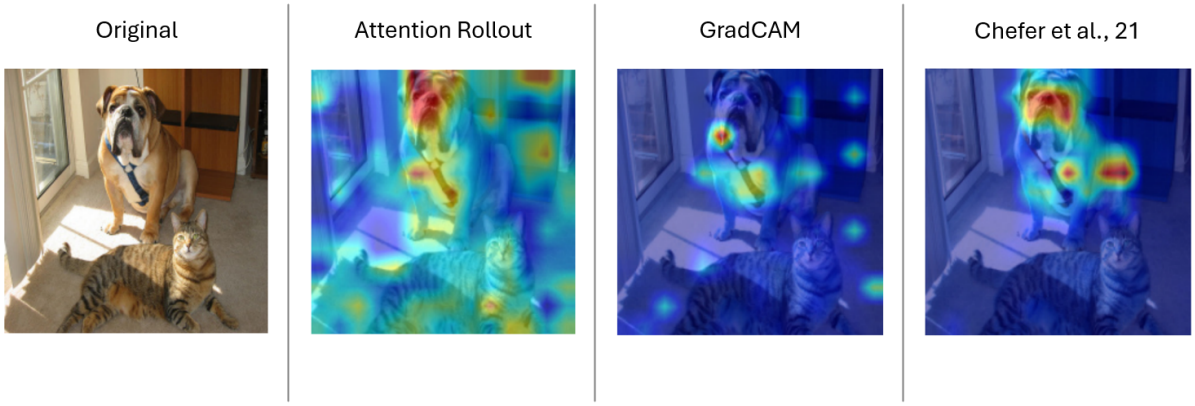


Source: Abnar and Zuidema, 2020 [3]

As discussed in the earlier paragraphs of this section, some XAI techniques are inherently model-specific, such as GradCAM, which relies on the (last) convolutional layer of a CNN. While applying the basic principles of GradCAM to a vision transformer such as ViT is technically feasible, potentially due to the fact that attention can be formally considered a special case of dimension-preserving convolution [17], these adaptations have not been proven equally effective. This limitation was demonstrated by Chefer et al., (2021) [11], who proposed an explanatory method based on deep Taylor decomposition that incorporates both

self-attention and skip connection components in vision transformer networks. In their main results, summarised in Figure 17, the authors demonstrated that their proposed method yielded superior performance compared to both the adapted version of GradCAM and standard attention rollout approaches.

Figure 17: Comparison of the outputs produced by different saliency map algorithms given the same input image for the classification label "`dog`".



Source: adapted from Chefer et al., 2021 b [11]

Both attention rollout and Chefer's methods produce attention-based saliency maps to understand where a model "pays attention" during the encoding process, typically by analysing the attention weights associated with the `CLS` token. While both methods can theoretically work with either vision or language transformers, they are limited to *single-modality* encoder-type models. However, as we observed at the end of Section 2.1, the field of medical AI has recently witnessed a surge in vision-language models (VLMs), which are generally fine-tuned from CLIP-based architectures. CLIP is a dual-encoder VLM constructed from pre-trained language and vision transformer encoders, such as BERT and ViT, respectively.

To explore how the two modalities interact within these dual encoders, Chefer and colleagues [10] proposed a method capable of extracting attention patterns from multiple types of dual-modal systems, including LXMERT [52] and CLIP. As illustrated in Figure 18, the proposed method enables researchers to understand which parts of both input modalities the model focuses on during the encoding process when given a VLM. In a medical context, this method could be particularly valuable when applied to scenarios such as feeding a model an X-ray scan paired with a clinical question, for example: "`Do you see a nodule? Is it benign or malignant?`". The method would then reveal which regions of the image the VLM focuses on and which specific terms, such as "benign" or "malignant", receive the most attention during the decision-making process. This capability is especially important for medical applications where understanding both visual and textual reasoning is crucial for building trust and ensuring clinical interpretability.

2.4 Mechanistic Interpretability Methods

As discussed in Section 1.1, the distinction between interpretability and explainability can often become blurred in practice. From a more formal theoretical perspective, most of the methods we have examined, such as GradCAM and attention rollout, should properly fall under the explainability definition. This is because they primarily provide *explanations* about the model's decision-making process—for instance, by highlighting the specific regions of the input that contribute most significantly toward a particular class

Figure 18: Multimodal saliency map for a VLM model.



Source: adapted from Chefer et al., 2021 a [10]

prediction. However, these same methods also offer us valuable *interpretations* of specific computational mechanisms operating within the model, such as the attention heads of transformers that are analysed in attention rollout techniques.

Mechanistic methods, on the other hand, can be more readily categorised under the interpretability umbrella, as they are specifically designed to precisely interpret very particular components of a model, typically transformer architectures. These components can range from highly granular elements, such as individual neurons within the MLP components, to broader structural elements, as exemplified by circuit analysis. In circuit analysis, researchers aim to understand—and potentially manipulate—entire subsets of a model that are collectively responsible for specific target behaviours or conceptual representations.

What distinguishes mechanistic tools as a specialised subset of interpretability methods is their commitment to maintaining a more rigorous connection to causal relationships, at least from a theoretical standpoint. This causal focus allows researchers to not only observe correlations but to make stronger claims about the underlying computational processes.

Finally, it is particularly noteworthy that while all the methods discussed thus far have predominantly presented results from more "traditional" predictive AI and ML—whether transformer or CNNs based—mechanistic methods have found their primary domain of application in decoder models, which are more commonly known as GenAI. This distinction reflects the different architectural priorities and interpretability challenges that arise when analysing models designed for generation versus those optimised for classification or representation learning.

2.4.1 Lenses

In Section 2.1.2, we explored how transformers can be categorised into three distinct architectural paradigms: encoder models like BERT, which are particularly well-suited for tasks such as classification; encoder-decoder models like T5, which excel in scenarios requiring sequence-to-sequence transformations such as translation and summarisation; and decoder-only models like GPT, which demonstrate superior performance in all aspects of text generation. Decoder-only models are also referred to as autoregressive models, meaning that given an input sequence of words—sometimes called a prompt—they generate text

incrementally, one token at a time. To accomplish this sequential generation, an autoregressive model processes the computational representation of the last word in a sequence (its embedding) produced by the final layer of the model, and maps this representation to the model's vocabulary through a linear transformation layer (illustrated in the upper right portion of Figure 4).

The fundamental insight behind Logit Lens [39] is to apply this same linear mapping layer not only to unveil predictions from the final layer, but systematically throughout the entire model architecture. This innovative approach enables us to interpret the internal decision-making processes of large language models (LLMs) by examining how the selection of the next predicted token evolves progressively across the model's various components and layers.

Figure 19 presents a concrete real-world example of Logit Lens application in practice. The X-axis displays the input tokens in the lower section (specifically, "`Specifically, we train GPT-3, an`"), while the predicted subsequent tokens appear in the upper section. The Y-axis indicates which specific components of the LLM—in this case, a GPT-2 model—are being queried and analysed. Each cell within the visualisation contains the predicted token for the corresponding component, while the colour intensity of the heatmap represents the numerical confidence score associated with each token prediction. In this particular example, we can observe how, in most instances, the model becomes increasingly confident in its token selection as computation progresses through later stages, regardless of whether the prediction is ultimately correct or incorrect.

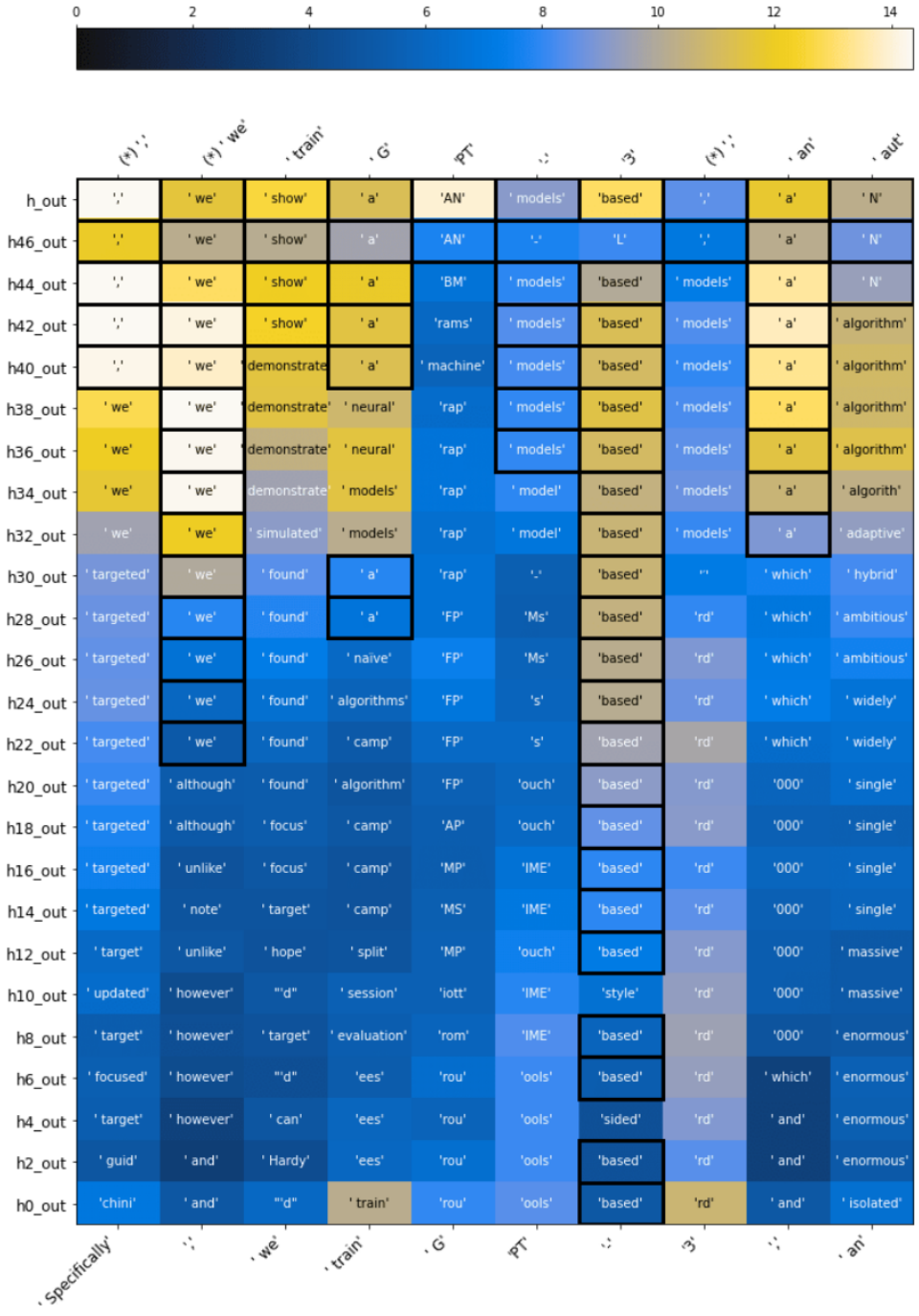
While the example illustrated in Figure 19 appears to produce largely sensible and interpretable results, other researchers have noted that Logit Lens can easily produce misinterpretations when applied to model components other than the final layer, and have consequently proposed alternative methodological solutions. One notable alternative approach is Tuned Lens [6], which involves training an affine probe specifically for each transformer block. As demonstrated in Figure 20, this refined solution can yield significantly more coherent and consistent interpretations, particularly for the earlier layers of the model where standard Logit Lens may provide misleading insights.

Both Logit Lens and Tuned Lens methodologies are specifically designed to work with generative LLMs, such as the various models in the GPT family. Recently, Joseph and colleagues [31] introduced Prisma, a mechanistic interpretability tool based on the core principles of Logit Lens, but adapted to interpret vision models such as Vision Transformers (ViT) and CLIP. Since ViT and CLIP are non-generative encoder models that operate fundamentally differently from language models, the core conceptual framework of Prisma differs substantially from both Logit and Tuned Lens approaches. Rather than analysing how predictions for subsequent tokens change across layers and token positions, Prisma investigates the connections between different image patches—the non-overlapping rectangular segments into which an input image is divided (as illustrated in Figure 6)—and the class labels in classification tasks, as seen in Figure 21.

To provide a brief recap of the underlying architecture, ViT begins its processing pipeline by systematically dividing each input image into non-overlapping rectangular patches or tiles. Using Prisma's interpretability framework, we can analyse and visualise which of the available classification classes each individual patch is most strongly connected to or associated with. In the illustrative example presented in Figure 21, the available classes are visually represented using intuitive emoji symbols for clarity. The central subfigure reveals how the correct classification of the "`cat`" class is primarily connected to a relatively limited set of centrally-located patches that indeed correspond to the region of the image containing the actual cat. Furthermore, we can observe how an even larger number of patches are coherently classified with alternative labels, providing insight into the model's decision-making process.

The question of how such a small subset of patches can effectively guide the model toward the correct classification solution is partially addressed in the rightmost subfigure of Figure 21. Here, we can intu-

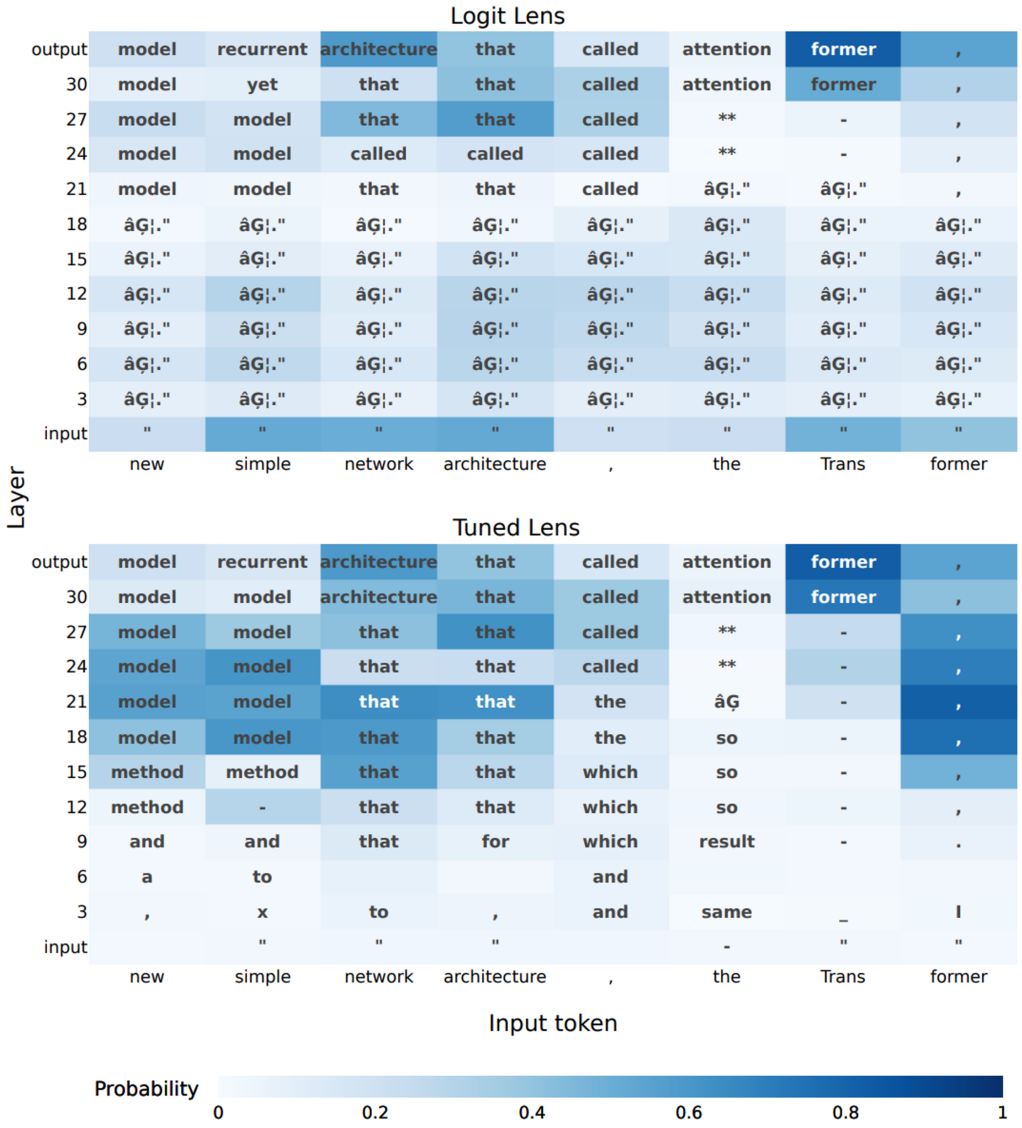
Figure 19: Logit Lens prediction across a GPT2 model for the input sentence "`Specifically, we train GPT-3, an`". The asterisk (*) on the top tokens indicates when the model's top guess matched the correct output. Outlined cells indicate that the choice at a given component matches the guess at the final layer.



Source: https://www.lesswrong.com/posts/AcKRB8wDpdaN6v6ru/interpreting-gpt-the-logit-lens

itively observe that the patches associated with the "`cat`" classification exhibit notably higher logit values compared to other regions. Moreover, we can see how, similar to the pattern observed in Figure 19 for the GPT-2 model using Logit Lens, the correct decision appears to emerge more clearly in the higher, more abstract layers of the model architecture.

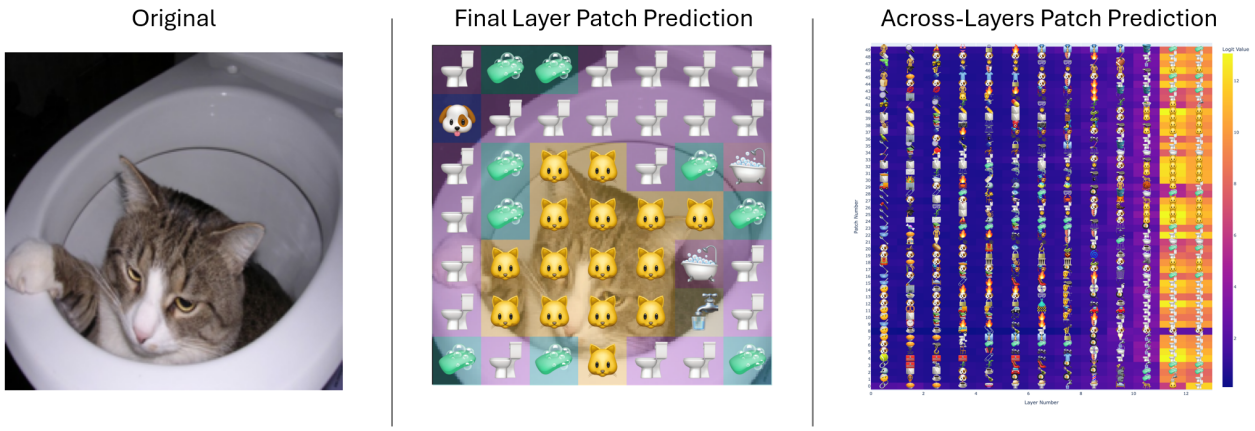
Figure 20: Comparison of Tuned Lens and Logit Lens outputs.



Source: adapted from Belrose et al., 2023 [6]

The potential applications of these interpretability tools to the medical field are both straightforward and promising. For instance, researchers could specialise a ViT model for specific medical imaging tasks such as pulmonary nodule detection or CVD classification. Subsequently, they could employ these interpretability techniques to understand not only which specific regions of the medical images are most strongly connected to particular diagnoses, but also whether other areas of the input contain meaningful diagnostic information or potentially problematic biased classifications. This latter point is particularly important, as it could help identify whether the model is inadvertently relying on demographic features such as age or biological sex rather than purely medical indicators, thereby helping to ensure more equitable and clinically sound diagnostic tools.

Figure 21: Original image and Prisma outputs. Emojis represent the classification label (e.g., ‘‘cat’’, ‘‘dog’’, ‘‘WC’’) predicted by the model for each path (i.e., image segment, see Figure 6).



Source: adapted from Joseph et al., 2025 [31]

2.4.2 Sparse autoencoders (SAE)

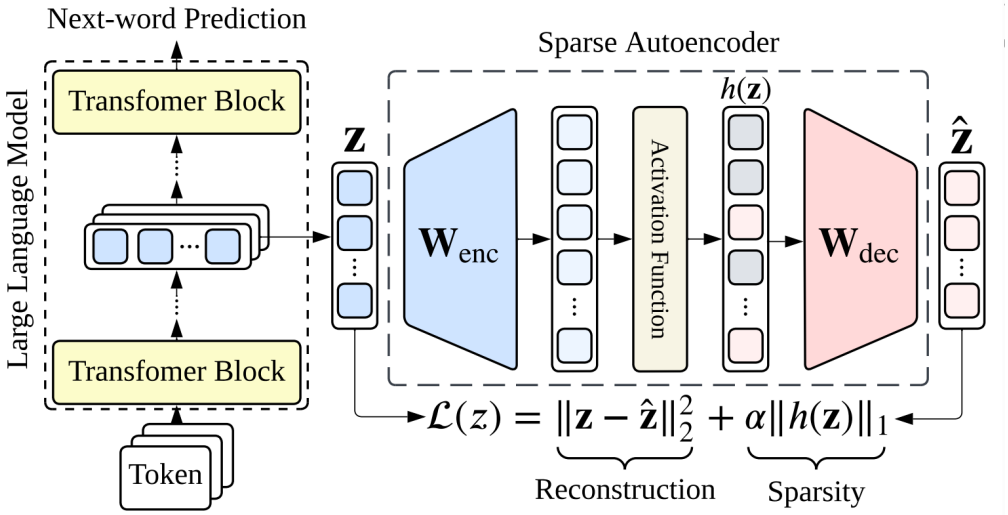
Sparse autoencoders (SAEs) represent a mechanistic interpretability method primarily designed to understand the internal workings of the representation produced by LLMs, specifically, the ones outputted by the feed-forward layer—also referred to as multi-layer perceptron (MLP). This approach emerged from the fundamental challenge of disentangling superimposed features within neural networks [50]. Extensive research indicates that neurons in LLMs frequently exhibit polysemantic behaviour, simultaneously encoding multiple distinct concepts within their activations—a phenomenon commonly referred to as superposition. This creates significant challenges for interpretability, as individual neurons cannot be easily mapped to specific conceptual meanings.

The theoretical foundation of SAEs rests on the linear representation hypothesis, which posits that neural networks encode concepts as specific directions within their high-dimensional activation spaces. Building upon this principle, SAEs are designed to learn sparse, linear, and decomposable representations of a model's internal activations. The key insight is that by projecting dense model representations onto a sparse vector space of higher dimensionality, individual features can be allocated to separate neurons, thereby enabling a clearer interpretation of the concepts that become active during model processing.

Figure 22 provides a visual representation of this approach. The process begins with model representations (denoted as Z), which are projected onto a sparse vector through matrix multiplication with an encoder matrix, followed by the application of an activation function. This transformation produces the SAE representation $h(z)$, which typically operates in a higher-dimensional space than the original model vector. Subsequently, this expanded representation is transformed back to the original dimensionality through multiplication with a decoder matrix, ensuring that the essential information content is preserved.

The training of SAEs relies on a carefully designed loss function that balances two competing objectives. The first component is a reconstruction loss, which ensures that the autoencoder can accurately reconstruct the original input from its sparse representation, maintaining fidelity to the original model activa-

Figure 22: Schematic diagram of generic SAE approach.



Source: Shu et al., 2025 [50]

tions. The second component is a sparsity loss, which actively encourages the model to produce sparse activations, thereby promoting the desired property of having few active features at any given time. Empirical research has demonstrated that SAEs successfully learn meaningful and interpretable features [54]. These learned features exhibit two crucial properties that validate their utility for mechanistic interpretability. First, they demonstrate specificity: when a particular feature becomes active, the corresponding concept is demonstrably present in the input context, establishing a reliable correspondence between feature activation and conceptual content. Second, these features exhibit causal influence on model behaviour: direct intervention on a feature's activation level produces predictable and relevant changes in the model's downstream processing and outputs, confirming that the identified features play genuine functional roles in the model's computation rather than merely representing spurious correlations.

2.4.3 Circuits

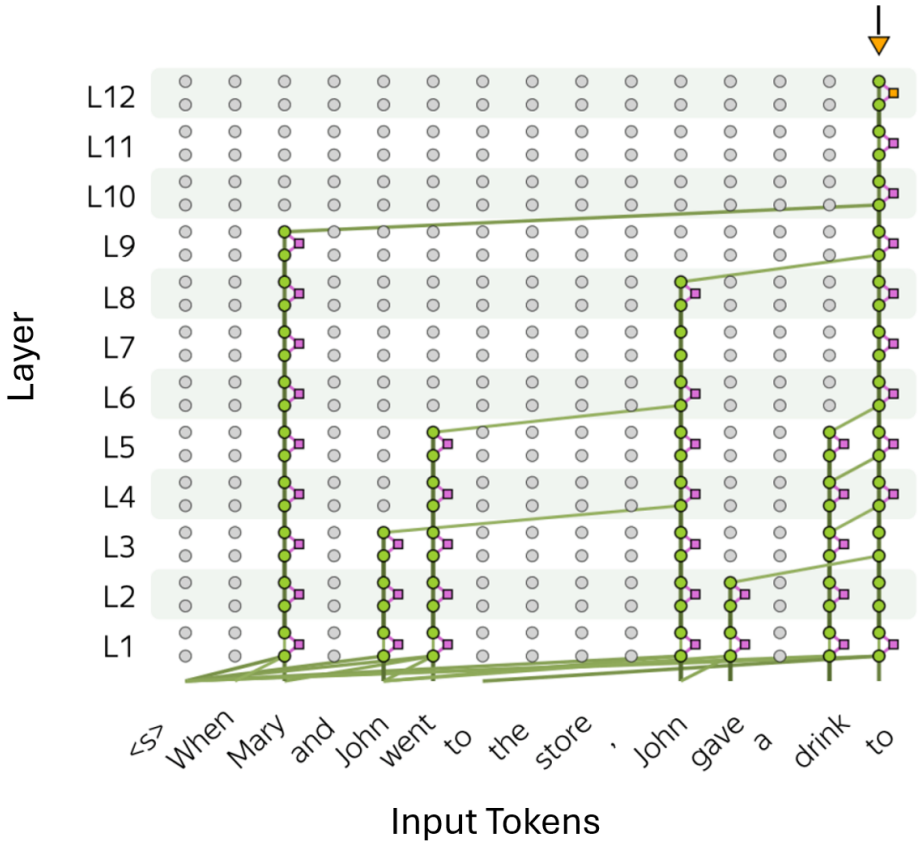
Throughout this section, we have systematically increased the granularity of our analytical focus. When examining lenses, our objective was to understand how predictions for a *single token* change across the outputs of different components. With SAEs, we broadened our scope of investigation, seeking to comprehend the *set of features* encoded within each neuron of a transformer's MLP. With circuit analysis, our goal becomes identifying a minimal subset of model components that collectively drive a specific *behaviour*.

From both theoretical and computational perspectives, transformers, like all artificial neural networks, can be conceptualised as computational graphs. In this framework, components such as MLPs or individual attention heads function as nodes, while activations—the outputs transmitted from one component to another—serve as the edges connecting these nodes. The fundamental principle underlying circuit analysis is to identify the minimal set of components, termed *a circuit*, that can account for specific model behaviours.

These behaviours encompass a diverse range of capabilities, from relatively simple tasks such as correctly predicting the next token [22] (as illustrated in Figure 23), to more complex linguistic phenomena

like indirect object identification [57]. Circuits have also been discovered for performing mathematical operations [28] and even for storing and retrieving precise factual knowledge [59]. This breadth of applications demonstrates the versatility of circuit analysis as a tool for understanding the internal mechanisms of transformer models.

Figure 23: Identified circuit that led the LLM (GPT2-small) to the predict the token ‘‘`Mary`’’.



Source: Ferrando et al., 2024 [22]

3 Challenges and Innovations in Explainable AI Applications for Healthcare

As highlighted in the introduction, explainable artificial intelligence (XAI) represents a crucial component of any AI-assisted or AI-augmented healthcare system. This importance has been thoroughly demonstrated in a recent comprehensive literature review that analysed more than 100 XAI scientific publications focusing specifically on medical applications [51]. The research demonstrates that XAI serves not only to ensure that healthcare practitioners fully understand the rationale behind a model's diagnostic or treatment recommendations but also to enhance patient involvement and deepen their understanding of their care pathways.

The review reveals that the most widely adopted techniques are perception-based or visually oriented methods, such as LIME, SHAP, and GradCAM, primarily due to their intuitive nature and simplicity of interpretation. Among these approaches, GradCAM appears to be the most frequently implemented method in medical imaging applications. However, this technique is inherently model-specific and can only be applied to CNNs. Although this architectural limitation might restrict the application of newer and more powerful models, such as Vision Transformers (ViTs) or CLIP, analogous methods have already been proposed and adapted for these advanced architectures, as discussed in Section 2.3.2.

The literature review emphasises the critical importance of multimodality for medical AI and healthcare XAI systems, arguing that multimodal approaches offer a more comprehensive understanding of patients' clinical presentations while simultaneously enhancing the overall user experience for healthcare professionals. However, the authors identified only 11 papers adopting multimodal approaches out of their review, highlighting the urgent need to expand research efforts in this promising area.

The review identifies five primary challenges within the XAI for healthcare literature that demand immediate attention and coordinated research efforts:

- **Data Quality and Standardisation:** The majority of current research relies heavily on publicly available datasets, which creates significant limitations in terms of quality standardisation and generates substantial inconsistencies across different studies. This fundamental issue directly affects the reliability, reproducibility, and comparability of XAI methods when applied to medical contexts. The lack of standardised evaluation metrics and benchmarks further compounds this problem, making it difficult to assess the true clinical utility of different XAI approaches.
- **Generalisation Limitations:** Due to the persistent lack of data standardisation and the inherent limitations of current medical AI systems to generalise effectively to new benchmarks and clinical environments, the generalisation capability of XAI methods remains a significant challenge. This limitation poses serious concerns for real-world deployment scenarios where models must perform consistently and reliably across diverse clinical settings, varying patient populations, different imaging equipment, and evolving medical protocols.
- **End-to-End Application Gap:** While the research community has extensively studied and validated most of the established XAI methods in controlled experimental settings, these approaches largely remain confined to the experimental stage in terms of actual deployment and seamless integration into comprehensive end-to-end clinical pipelines. This substantial gap between theoretical research achievements and practical clinical application represents one of the most significant barriers to the widespread clinical adoption of XAI technologies.
- **Bias Detection Inadequacy:** A growing body of evidence has consistently demonstrated how both general-purpose and medical AI systems can perpetuate and amplify harmful demographic biases, yet there appears to be insufficient literature addressing how XAI methodologies could contribute

to detecting, understanding, and mitigating these biases. Experimental results conducted within our research unit indicate that while some XAI methods can provide a reasonable degree of detection capability for relatively explicit biases, they may struggle significantly to identify more subtle demographic biases, particularly those based on biological sex, race, or socioeconomic factors.

— **Implementation Barriers:** The literature has consistently shown that while most XAI methods have been extensively validated and proven effective in other domains and applications, their practical implementation in real-world medical settings remains severely limited. This limitation likely stems from a complex interplay of the previously mentioned challenges, regulatory requirements, clinical workflow integration difficulties, and the conservative nature of healthcare institutions, creating an intricate web of barriers that must be addressed holistically to advance the field toward meaningful clinical impact.

These interconnected challenges can simultaneously encompass technical, organisational, and regulatory aspects, as suggested in Table 1. Collectively, these challenges demonstrate that while XAI holds tremendous promise to improve the transparency, trustworthiness, and effectiveness of healthcare AI systems, substantial coordinated research and development efforts are necessary to bridge the existing gap between theoretical capabilities and practical clinical implementation. Addressing these challenges will require collaborative efforts among AI researchers, healthcare professionals, regulatory bodies, and technology developers to ensure that XAI systems can deliver on their potential to transform healthcare delivery while maintaining the highest standards of patient safety and clinical efficacy. Indeed, some of these efforts are already underway and having a concrete impact. For instance, data quality and standardisation are critically relevant for the healthcare domain, both for AI model training and privacy considerations, especially given AI systems' potential to analyse vast amounts of data that often remain unutilised [15]. In this space, recent years have seen the rise of multiple European initiatives, such as the European Health Data Space (EHDS)[1] and the European Federation for Cancer Images (EUCAIM)[2], as extensively described in a recently published JRC report [16].

Table 1: Challenges and limitations in AI for health

Challenges	**Type**		
	Technical	**Organisational**	**Regulatory**
Data Quality and Standardisation		X	X
Generalisation Limitations	X		
End-to-End Application Gap		X	X
Bias Detection Inadequacy	X		X
Implementation Barriers	X	X	

Source: own elaboration

[1] https://www.european-health-data-space.com/
[2] https://dashboard.eucaim.cancerimage.eu/

4 Use cases in Cardiovascular Diseases

4.1 State of the Art

Reading the review discussed in the previous sections, we can see how medical researchers in XAI have devoted substantial efforts to studying diseases affecting the respiratory system, including pneumonia, lung cancer and, more recently, COVID-19. Another anatomical region that has garnered considerable attention from the same research community is the cardiovascular system, as CVDs currently represent one of the leading causes of mortality worldwide [45].

A recent study reviewed approximately 200 scientific publications focusing on XAI approaches for cardiac research, utilising multiple databases including Web of Science, Scopus, PubMed, and IEEE [46]. Given the complexity of tracking and systematically analysing every result obtained using XAI methods within a traditional scientific paper format, this investigation primarily concentrated on three key dimensions of the extracted works: the type of data (or modalities) employed in the experiments, the specific XAI tools that were adopted, and the nature of the output these tools produced in terms of alignment with human expectations.

Concerning the data modalities, the review highlighted how the literature has investigated a large number of possible solutions, encompassing most of the existing data modalities relevant to cardiac science. These include textual data from electronic health records (EHR), time series data such as electrocardiograms (ECG), medical imaging data including X-rays, Magnetic Resonance Imaging (MRI) and ultrasound, as well as tabular data from laboratory tests and clinical measurements. This coverage demonstrates the field's commitment to leveraging diverse data sources for cardiovascular AI applications. As illustrated in Figure 24, GradCAM and SHAP emerge as by far the most widely adopted explainability methods across these various modalities.

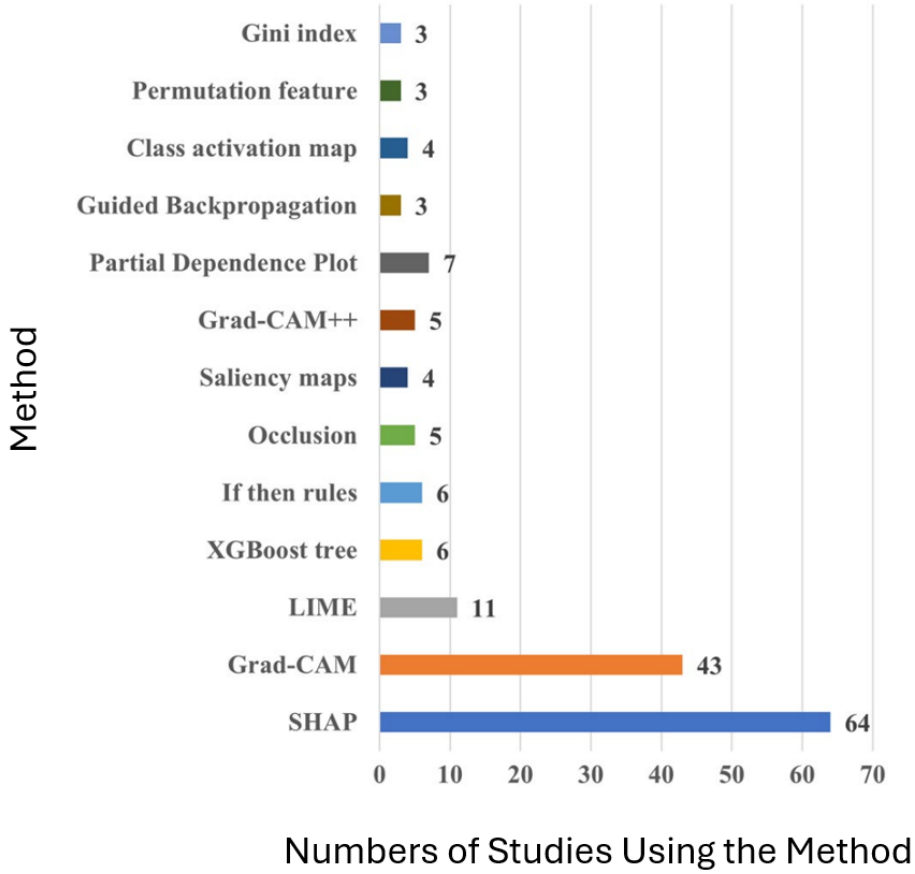
Another significant finding from the literature review is that the vast majority of papers under investigation reported a positive alignment between XAI tools and expert knowledge. Specifically, the explanations and outcomes produced by these tools matched the expectations and clinical knowledge of field experts, as demonstrated in Figure 25. This high rate of expert-AI alignment suggests promising potential for clinical translation of these explainable AI systems. Unfortunately, the work does not provide detailed insight into how these matches are distributed across the different XAI tools and data modalities, which would be valuable in providing methodological recommendations and guiding future research directions in the field.

The literature review encompassed publications through August 2023. Subsequently, the research community has continued to advance novel AI architectures and pre-trained models, with particular emphasis on general-purpose and medical vision-language models (VLMs). Despite the growing prominence of this research direction, its influence on the specialised field of explainable artificial intelligence (XAI) applications in cardiac science remains limited.

For instance, Duyar and colleagues [20] developed a deep neural network model utilising a one-dimensional CNN (1D-CNN) architecture to detect cardiovascular disease through the analysis of bacterial taxonomy and operational taxonomic unit (OTU) data. Their approach was benchmarked against established ML models commonly employed for tabular data analysis, including support vector machines (SVM), XGBoost, and TabNet, with the proposed 1D-CNN demonstrating superior classification performance across all evaluated metrics.

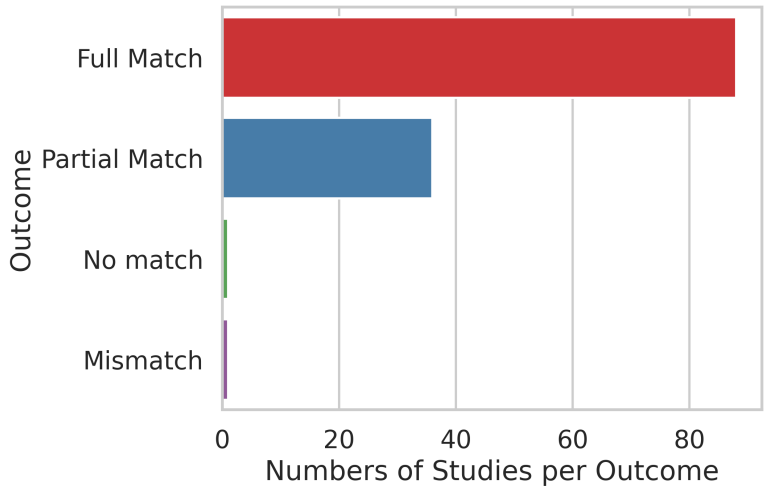
The 1D-CNN model achieved the highest area under the curve (AUC) value of 97.09% when processing bacterial taxonomy data, notably accomplishing this result despite working with a smaller-than-anticipated sample size. This impressive performance metric underscores the model's robustness and effectiveness in

Figure 24: Distribution of XAI methods adopted in cardiology-based literature.



Source: Salih et al., 2024 [46]

Figure 25: Distribution of XAI expectation outcomes between XAI explanation and field experts in cardiology-based.



Source: adapted from Salih et al., 2024 [46]

distinguishing between CVDs cases and healthy controls using microbiome-derived features. The achievement is particularly noteworthy given the inherent challenges associated with limited sample sizes in medical datasets, which often lead to overfitting and reduced generalisability in deep learning applica-

tions.

To identify bacterial taxa most strongly associated with cardiovascular disease, the research team employed multiple complementary analytical approaches, including Pearson's correlation analysis, LASSO regularisation, and SHAP values as an explainable artificial intelligence technique. From their comprehensive analysis of 455 bacterial species, they systematically focused on the top 20 most significant contributors to disease classification, providing a manageable subset for detailed biological interpretation.

The analytical methods revealed intriguing and concerning discrepancies in their findings, raising important questions about feature importance consistency across different XAI approaches. Both Pearson's correlation and LASSO regularisation consistently identified `k_Bacteria;p_Tenericutes` as the most highly correlated taxonomic group with cardiovascular disease, suggesting a strong linear relationship between this bacterial phylum and disease presence. However, the SHAP analysis diverged substantially from these results, instead designating `k_Archaea;p_Crenarchaeota` as the primary taxonomy associated with the condition, while relegating `k_Bacteria;p_Tenericutes` to the 17th position in terms of feature importance rankings.

These methodological inconsistencies highlight the inherent complexity of identifying discriminative taxonomic features in microbiome-based disease prediction and underscore a critical limitation in current XAI methodologies. The divergent results suggest that different analytical approaches may capture distinct aspects of the complex microbiome-cardiovascular disease relationship, potentially reflecting differences in how linear correlations, regularised feature selection, and game-theoretic attribution methods interpret high-dimensional biological data. These findings warrant further investigation to reconcile these divergent results and establish more robust consensus approaches for feature importance assessment in cardiovascular XAI applications.

SHAP values were also adopted by researchers who developed a comprehensive approach to improve heart disease detection by implementing a stacked ensemble model that integrates multiple ML techniques [25]. Their methodology involved selecting the most effective algorithms—including Random Forest, MLP, K-Nearest Neighbours (KNN), Extra Tree Classifier, XGBoost, Support Vector Classifier (SVC), Stochastic Gradient Descent (SGD), AdaBoost, Classification and Regression Trees (CART), and Gradient Boosting Machine (GBM)—based on their individual performance with raw data.

To address the inherent challenges of feature selection in medical data, the team employed three distinct feature selection strategies: Pearson correlation analysis, Chi-Square testing, and Recursive Feature Elimination (RFE). This multipronged approach enabled the model to concentrate on the most diagnostically relevant features while filtering out potentially confounding variables. The proposed ensemble model achieved remarkable results, demonstrating an accuracy rate of 93.3%. This performance metric suggests significant potential for clinical application in cardiovascular risk assessment.

The SHAP value analysis revealed that several clinical variables exhibited a strong predictive influence: exercise-induced angina, ST slope characteristics (both upslope and flat patterns), and male sex emerged as significant positive contributing factors to the prediction of heart disease. Interestingly, the maximum heart rate achieved during testing showed a complex relationship with disease prediction—higher values contributed negatively to disease probability, while lower values showed a moderately positive association with cardiac issues.

More recently, a study [24] has focused on enhancing heart disease prediction using stacking and voting ensemble methods, using SHAP to interpret the model's predictions. The researchers trained fifteen base models on two heart disease datasets. After evaluating various combinations, they selected six base

models to develop ensemble approaches employing a meta-model (stacking) and majority voting. The performance of these ensemble models was compared against individual base models.

Results indicated that the ensemble models, particularly stacking, consistently outperformed individual models, achieving higher accuracy and improved predictive outcomes. The researchers incorporated XAI through SHAP analysis to interpret model predictions.

In the stacking model applied to the first dataset (D1), the slope of the peak exercise ST segment (STS) emerged as the most significant feature, reflecting its strong correlation with heart disease. Chest pain type (CP) proved critical, consistent with its established diagnostic relevance. Other influential features included serum cholesterol (CL) and gender (GD), which contributed moderately to predictive power. Conversely, features like fasting blood sugar (FBS), resting electrocardiographic results (REC), and maximum heart rate achieved (MHR) showed limited impact, likely due to weaker associations with heart disease in this dataset.

For the second dataset (D2), the stacking model identified the number of major vessels coloured by fluoroscopy (CF) as the dominant predictor, indicating its importance in distinguishing heart disease within this population. CP retained high SHAP values, emphasising its relevance across datasets. Features such as thallium heart rate (TH) and oldpeak (OP) also gained prominence in D2, reflecting their increased significance in this dataset's patient demographics. These differences suggest that dataset-specific attributes influence model behaviour.

4.2 Alternative XAI Methods

Indeed, there are some exceptions to the list of methods presented in the review. For instance, researchers have tried to combine and test which solution, between integrated gradients or layer-wise relevance propagation (LRP), could replace GradCAM [7]. This work applies explainable attribution methods to a pre-trained deep neural network for abnormality classification in 12-lead electrocardiography to understand the relationship between model predictions and learned features.

XAI for time series data, such as biosignals, requires further research. Available XAI methods for deep neural networks include layer-wise relevance propagation (LRP), integrated gradients (IG), and GRAD-CAM. However, for ECG classification, these methods are typically applied qualitatively by displaying individual recordings with corresponding XAI information as pseudo-colored overlays. This qualitative evaluation of single recordings provides only anecdotal evidence and fails to meet the requirements for integrating deep neural networks in clinical practice, which demands comprehensive model characterisation and limitation assessment.

The authors evaluate the explanatory power of different XAI methods across two databases. These XAI methods assign relevance scores to each ECG time series sample, reflecting its influence on the network's decision. The main contribution involves novel analysis methods for processing these scores, enabling insight into the network's reasoning when classifying unseen ECG signals. By mapping results to clinical knowledge, the researchers investigate how well the network's features align with established medical understanding while proposing novel visualisation methods for intuitive assessment of network classifications.

The analysis suggests that the network learned features similar to those in cardiology textbook knowledge. The results indicate that both the IG and LRP methods effectively provide insight into deep neural network reasoning for biosignals. Additionally, the comparison between IG and LRP methods revealed that IG produces the most distinct results.

In a recent paper [29], authors have examined ageing effects in the electrocardiogram (ECG) as a novel biomarker for cardiovascular risk, including atrial fibrillation (AF), heart failure (HF), myocardial infarction (MI), and mortality. To determine which ageing affects the AI-ECG model learned across all subjects, leads, and waveforms, the authors extended an established XAI workflow based on integrated gradients.

During ageing, two main adaptations become visible on the ECG. The first is ventricular hypertrophy, which describes cardiac remodelling necessary to compensate for the stiffness of aged vessels, especially of the aortic arch. The second is fibrosis, the molecular consequence of remodelling in which functional tissue is replaced by connective tissue to limit dilatation by holding the enlarged myocardium together.

This work adopts an AI-ECG model based on a ResNet architecture, pre-trained on clinical ECG exams from the Brazilian TeleHealth Network of Minas Gerais. The model processes 12-lead ECGs and predicts patient age in an end-to-end fashion. To explain the model's predictions, the authors use the post-hoc attribution method, integrated gradients to gain insight into AI-ECG decision-making. They compute "relevance" values for each input sample of each processed ECG and visualise them on top of the standard 12-lead ECG to highlight the most relevant parts for ageing effects.

The results indicate that the AI-ECG model associates ageing effects with increased mortality risk, given by an HR of 1.41 (1.27–1.57, $p < 0.001$) for "Overestimation" (more pronounced ageing effects) over the 21-year follow-up period of the SHIP study.

The AI-ECG model captures essential age-related cardiac changes applicable across various locations, healthcare systems, and patient populations, as the ECGs were acquired from different cohorts (USA, Brazil, Germany) and stem from population and clinical studies. While previous studies have developed AI models that directly predict individual mortality from ECG data, this indirect approach—using ageing effects as a cardiovascular biomarker—associates not only with mortality but also with the incidence of specific CVDs, such as AF and HF. This indicates that ageing effects may reflect subclinical pathologies at an early stage. The XAI analysis of individual ECG signals for their relevance distributions clearly demonstrates that relevances aggregate at regular positions across the P-QRS-T patterns.

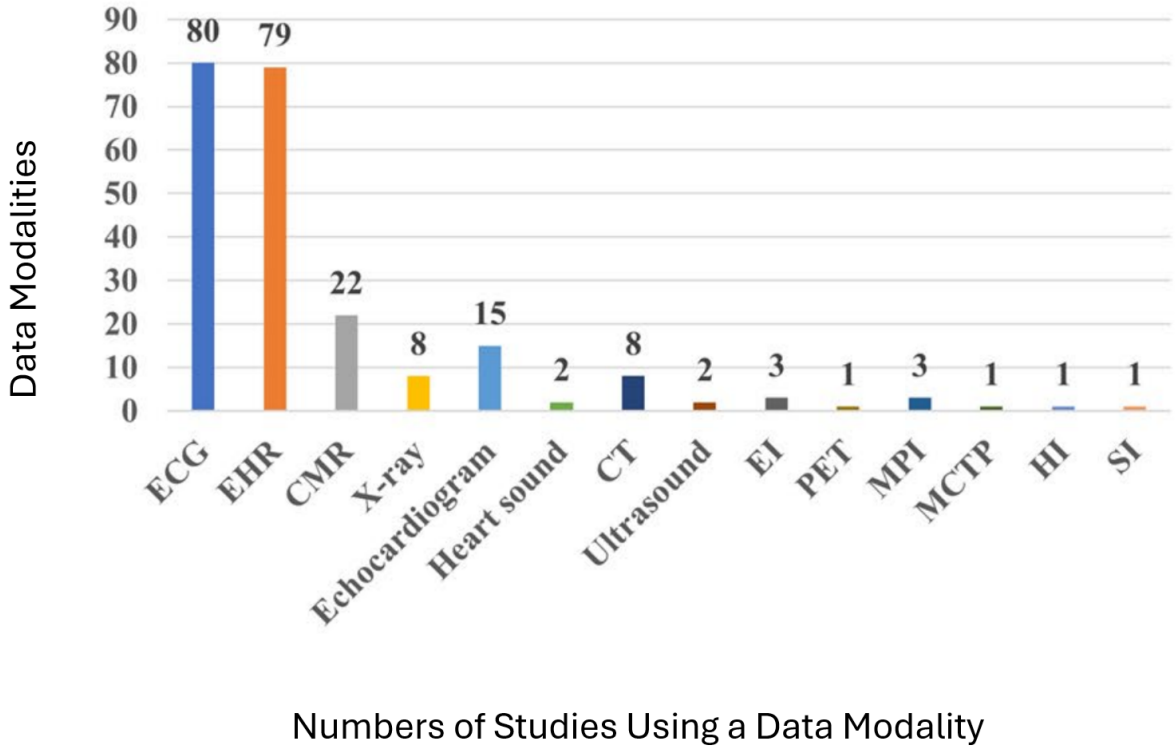
4.3 Research gaps and limitations

Overall, the review and presented individual studies highlight two significant limitations in the field of XAI for cardiac and cardiovascular (CVD) research. The first limitation is the predominant focus on unimodality. As demonstrated by Figure 26, the vast majority of studies under investigation concentrated on two specific modalities: ECG and EHR data. Crucially, there appears to be a systematic absence of studies that consider multiple modalities in combination.

The limited presence of multimodal approaches is not only evident from the figure but also from the textual analysis of the reviewed literature. Throughout the body of the review, the term "multimodal" is not mentioned even once. Furthermore, the bibliography contains only two papers with this word in the title, both belonging to rather recent publications. Finally, it is worth noticing how the general review discussed at the introduction of this section, which consists of a whole chapter on multimodal approaches to XAI for health, does not mention any application of multimodal XAI for cardiac or CVD applications.

This overwhelming preference for unimodal solutions represents a significant limitation in the current research landscape. The integration of multiple modalities has been repeatedly demonstrated to be beneficial in various domains, as it allows models to merge complementary information that may be modality-specific [51]. This approach can capture a more comprehensive picture of patient health status by taking advantage of the unique strengths of different data types. Moreover, EHRs are frequently text-based and

Figure 26: Distribution of data modalities investigated in XAI in cardiology-based literature.



Source: Salih et al., 2024 [46]

could therefore be readily encoded using pre-trained large language models (LLMs). These encoded representations could then be seamlessly integrated with modality-specific ECG models without requiring additional training of the ECG model to adapt to the text embeddings. This presents an opportunity for more sophisticated and comprehensive cardiac AI systems that could improve diagnostic accuracy and the interpretability of AI-driven clinical decisions.

The second limitation that emerges from the review is that research has been largely constrained to models like logistic regression, deep neural networks (DNNs), and convolutional neural networks (CNNs). While these approaches offer computational efficiency and have proven effective for specific tasks, they typically lack more sophisticated reasoning and contextual understanding capabilities, necessary for handling complex multimodal scenarios that require the integration of diverse information types. This limitation becomes particularly pronounced when tasks demand nuanced interpretation of relationships between visual and textual elements, or when complex inference across multiple modalities is required.

Both of these limitations can be effectively addressed through the adoption of Vision-Language Models (VLMs), which have gained significant prominence in recent literature due to their enhanced architectural sophistication. VLMs represent a paradigm shift in multimodal processing, offering advanced capabilities to simultaneously process, analyse, and understand visual and textual information through unified representational frameworks. In the CVD framework, the advantage of using these systems further lies in the capability of VLMs to handle multiple types of data modalities, such as time series or echocardiogram data, through the image encoder of VLMs. Indeed, as mentioned earlier in the report, there are already a few instances of successful application of VLMs in medicine. Qilt-Net is a VLM based on CLIP designed specifically for histopathology [30]. UniMedCLI was recently introduced as the first medical VLM

encompassing multiple types of medical domains and modalities [32]. These research efforts also include EchoCLIP [13], a VLM trained on echocardiogram and text data, that has shown significant potential in assessing cardiac function and identifying implanted intracardiac devices.

A limitation that was not directly pointed out in the review, but that seems to emerge from its content, is how integrated the examined methods were into actual clinical workflows, or whether each and every study under consideration was reporting findings obtained in a research setting. This difference surely does not impact relevant findings such as the strong alignment between expert expectations and XAI outputs, but makes it harder to grasp how ready each tool is to be integrated into existing pipelines, or even as a stand-alone tool for everyday practice.

Before concluding, it is worth mentioning that the review [46] also points out the lack of causality addressed by these methods. Indeed, causality is generally not addressed by XAI methods, but it is of central interest in the mechanistic area of interpretability research. Generally speaking, interpretability has been largely understudied in medical AI. As we have seen, the focus has largely been on explainability, both for general medical AI and in the case of CVD applications. However, there are a few specific instances of work going in this direction, applying SAEs to healthcare tasks. Although there appears to be no application specifically to CVD, these studies indicate that using SAEs for CVD could lead to more interpretable models. One research study used SAEs for a multimodal model trained to produce radiology reports from chest radiographs [2]. The research showed that SAEs can be successfully implemented on multimodal tasks and can provide interpretable visual concepts and reports. The researchers used a Gated SAE architecture based on [42]. Using this architecture, they collected feature activations for radiology images from a vision encoder. To interpret and convert these features into radiology reports, they used an automated interpretability technique to analyse ground truth radiology reports of the highest activating images for each feature. Their findings are further supported by another study using SAEs for pathological images [34]. While single embedding dimensions showed polysemanticity, through the implementation of SAEs, the researchers could extract interpretable features.

5 Conclusions

This report examines the evolving landscape of explainability and interpretability tools alongside the AI models most frequently adopted by researchers and practitioners. The primary objective is to understand the capabilities and limitations of explainable AI (XAI) tools in revealing the inner workings, decision-making processes, and prediction mechanisms of complex AI systems, with particular attention to their contributions in enhancing trustworthiness and transparency of AI outcomes for decision-making processes.

The analysis encompasses both theoretical foundations and practical implementations of XAI methodologies, evaluating their effectiveness across different model architectures and problem domains. By systematically reviewing the current state of explainability techniques, this report aims to bridge the gap between technical innovation and real-world applicability, with a specific focus on healthcare applications where interpretability is paramount.

Healthcare presents unique challenges for AI interpretability, given the critical nature of medical decision-making and the high stakes involved in diagnostic processes. This report provides an in-depth analysis of how explainability tools and AI models have been strategically implemented in research initiatives and clinical applications within the health and cardiovascular domains. Through detailed case studies and empirical evidence, we examine the practical impact of XAI tools in improving diagnostic accuracy, enhancing clinical workflows, and building trust between healthcare professionals and AI-assisted systems.

The cardiovascular disease (CVD) domain deserves particular attention due to the complexity of physiological data and the critical nature of diagnostic decisions, which necessitate robust explainability frameworks. By analysing real-world implementations, we provide insights into best practices, common challenges, and emerging trends in the application of explainable AI within these critical healthcare domains.

Our literature review reveals a consistent predominance of two computational approaches to data modelling in CVD applications. The first encompasses an ensemble of “traditional” machine learning (ML) algorithms, including logistic regression and deep learning (DL) methods. The second focuses on convolutional neural networks (CNNs), which have proven particularly effective for medical imaging applications.

Regarding XAI tools, a recent and exhaustive review [46] has clearly demonstrated how XAI applications for CVDs have predominantly relied on two approaches: SHAP values and GradCAM. These methods have been extensively tested and widely integrated within both research and practice communities, representing lightweight, effective, and straightforward solutions that have proven their value in numerous applications.

However, the AI research community has increasingly shifted towards more powerful approaches, largely based on large language models. Among these advanced approaches, health and medicine-oriented applications have demonstrated strong interest in Vision-Language Models (VLMs) [32, 30], including pioneering efforts specifically designed for CVDs [13]. These models are particularly appealing as they combine relative computational efficiency with powerful classification capabilities—the primary focus in health and CVD domains.

This significant advancement in computational power naturally introduces increased complexity costs. In parallel, the XAI research community has responded by producing a vast array of sophisticated tools designed to understand these black-box models and their inner mechanisms. These encompass both established methods like SHAP and more specialised approaches, including Sparse Autoencoders (SAE) and concept-based methods that offer deeper insights into model behaviour.

We argue that the community of practitioners should rapidly adapt to these "Swiss-army-knife" models, particularly specialised VLMs paired with advanced XAI tools specifically designed for these architectures. The convergence of powerful VLMs with sophisticated XAI methodologies represents a critical opportunity to advance both the effectiveness and interpretability of AI systems in healthcare applications.

Furthermore, we contend that research efforts from interested communities should be strategically focused on advancing several key areas: boosting VLM generalisation capabilities, implementing robust bias mitigation strategies, and enhancing the power of concept-based tools capable of unveiling complex multimodal issues. These priorities are intrinsically tied to energy efficiency considerations, as more capable and interpretable models can potentially reduce computational overhead while maintaining or improving performance.

Together with these technical reconsolidations, we underline the support for real-world validation in clinical settings, which is essential for establishing trustworthy AI systems. This is particularly critical for cardiovascular disease (CVD) risk models that may directly influence patient care decisions. In such a setting, future XAI research should also evaluate practical usability by clinicians, including factors such as interpretability speed, user interface design, and seamless integration with electronic health records (EHR). Additionally, research must assess patient understanding and acceptance of AI-generated explanations.

Ultimately, this strategic alignment between advanced modelling capabilities and sophisticated interpretability tools will lead to more trustworthy and clinically relevant solutions, better serving the critical needs of healthcare practitioners and patients alike.

Box 1: Schematic summary of the overall report.

— **Objective**: Examines XAI tools and AI models to understand how they reveal AI decision-making processes to enhance transparency in healthcare applications, with a focus on CVD.

— **Main Findings:**

- Current State: CVD applications predominantly use traditional ML algorithms and CNNs, with XAI tools relying mainly on SHAP values and GradCAM as proven, lightweight solutions.
- Model Shift: AI research has shifted toward more recent architectures, such as Large Language Models (LLMs), and Vision-Language Models (VLMs) that offer powerful classification with computational efficiency.
- New tools uptake: LLMs and VLMs are more complex, but offer access to adapted versions of established methods (e.g., SHAPE, GradCAM), and more sophisticated tools like multimodal concept-based methods, Sparse Autoencoders (SAE), or Circuits.

— **Strategic Recommendations**: Research efforts should be increased in specific directions, such as specialised VLMs with advanced XAI tools, generalisation, bias mitigation, and concept-based methods for better clinical solutions.

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